Code

EORTC HNSCC ICI Honore et al_02032025

library(swimplot) library(grid) library(gtable) library(readr) library(mosaic) library(dplyr) library(survival) library(survminer) library(ggplot2) library(scales) library(coxphf) library(ggthemes) library(tidyverse) library(gtsummary) library(flextable) library(parameters) library(car) library(ComplexHeatmap) library(tidyverse) library(readxl) library(janitor) library(DT) library(pROC) library(rms)

#ctDNA Detection Rates by Window and Stages

Hide

```
#ctDNA at Baseline
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data$ctDNA.Base <- factor(circ_data$ctDNA.Base, levels=c("NEGATIVE","POSITIVE"))
circ_data <- subset(circ_data, ctDNA.Base %in% c("NEGATIVE", "POSITIVE"))
circ_data$Stage <- factor(circ_data$Stage, levels=c("0","I","II","III","IV"))
positive_counts_by_stage <- aggregate(circ_data$ctDNA.Base == "POSITIVE", by=list(circ_d
ata$Stage), FUN=sum)
total_counts_by_stage <- aggregate(circ_data$ctDNA.Base, by=list(circ_data$Stage), FUN=l
ength)
combined_data <- data.frame(
  Stage = total_counts_by_stage$Group.1,
  Total_Count = total_counts_by_stage$x,
  Positive_Count = positive_counts_by_stage$x,
  Rate = (positive_counts_by_stage$x / total_counts_by_stage$x) * 100  # Convert to perc
entage
)
combined_data$Rate <- sprintf("%.2f%%", combined_data$Rate)
overall_total_count <- nrow(circ_data)
overall_positive_count <- nrow(circ_data[circ_data$ctDNA.Base == "POSITIVE",])
overall_positivity_rate <- (overall_positive_count / overall_total_count) * 100  # Conve
rt to percentage
overall_row <- data.frame(
  Stage = "Overall",
  Total_Count = overall_total_count,
  Positive_Count = overall_positive_count,
  Rate = sprintf("%.2f%%", overall_positivity_rate)
)
combined_data <- rbind(combined_data, overall_row)
print(combined_data)
```

Stage <fctr>	Total_Count <int>	Positive_Count <int>	Rate <chr>
I	1	1	100.00%
II	5	4	80.00%
III	3	3	100.00%
IV	19	17	89.47%
Overall	29	25	86.21%

5 rows

Hide

```
#ctDNA post-treatment
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.postTx!="",]
circ_data$ctDNA.postTx <- factor(circ_data$ctDNA.postTx, levels=c("NEGATIVE","POSITIV
E"))
circ_data$Stage <- factor(circ_data$Stage, levels=c("0","I","II","III","IV"))
positive_counts_by_stage <- aggregate(circ_data$ctDNA.postTx == "POSITIVE", by=list(circ
_data$Stage), FUN=sum)
total_counts_by_stage <- aggregate(circ_data$ctDNA.postTx, by=list(circ_data$Stage), FUN
=length)
combined_data <- data.frame(
  Stage = total_counts_by_stage$Group.1,
  Total_Count = total_counts_by_stage$x,
  Positive_Count = positive_counts_by_stage$x,
  Rate = (positive_counts_by_stage$x / total_counts_by_stage$x) * 100  # Convert to perc
entage
)
combined_data$Rate <- sprintf("%.2f%%", combined_data$Rate)
overall_total_count <- nrow(circ_data)
overall_positive_count <- nrow(circ_data[circ_data$ctDNA.postTx == "POSITIVE",])
overall_positivity_rate <- (overall_positive_count / overall_total_count) * 100  # Conve
rt to percentage
overall_row <- data.frame(
  Stage = "Overall",
  Total_Count = overall_total_count,
  Positive_Count = overall_positive_count,
  Rate = sprintf("%.2f%%", overall_positivity_rate)
)
combined_data <- rbind(combined_data, overall_row)
print(combined_data)
```

Stage <fctr>	Total_Count <int>	Positive_Count <int>	Rate <chr>
I	1	1	100.00%
II	5	1	20.00%
III	3	2	66.67%
IV	19	15	78.95%
Overall	29	20	68.97%

5 rows

#Overview plot

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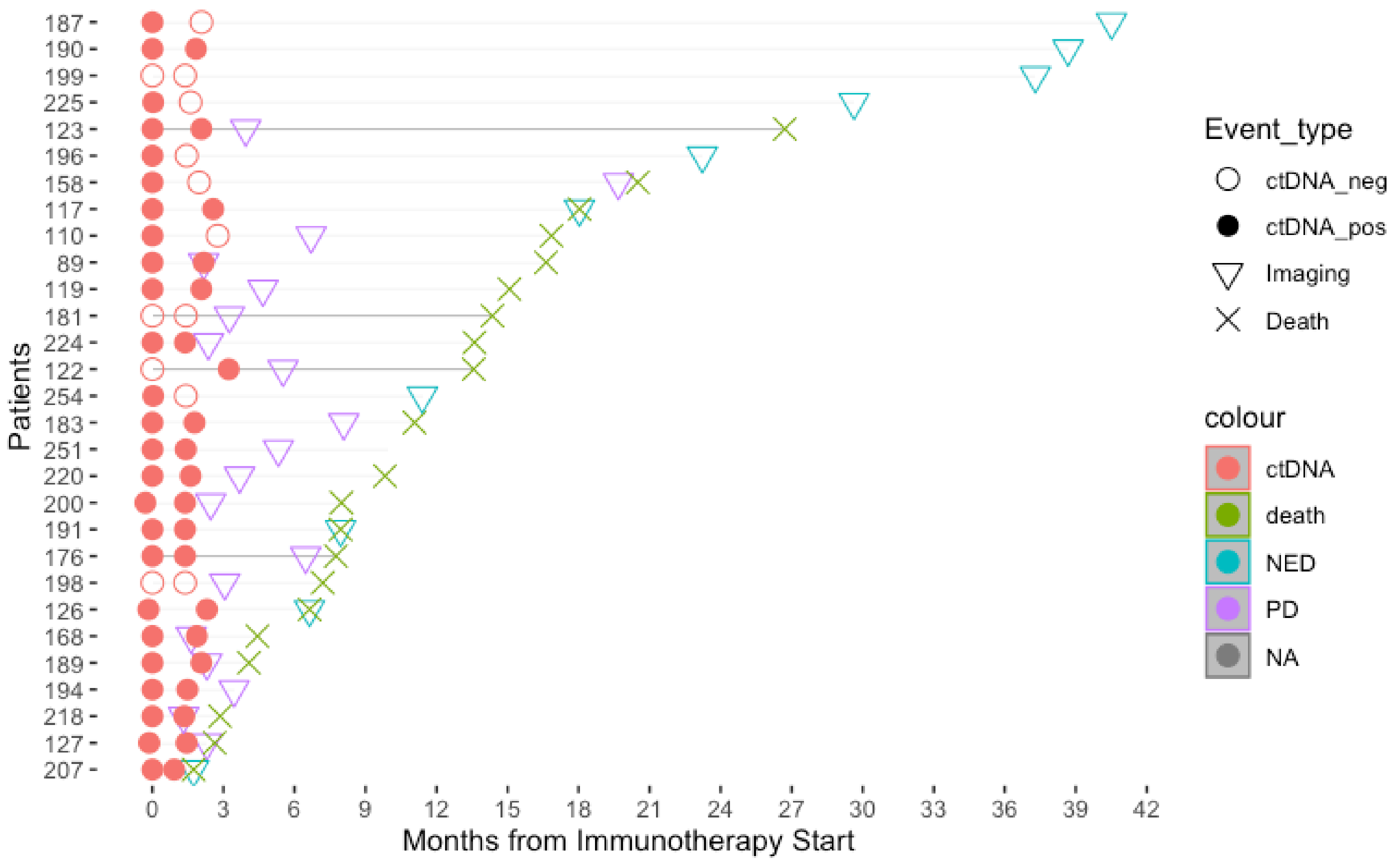
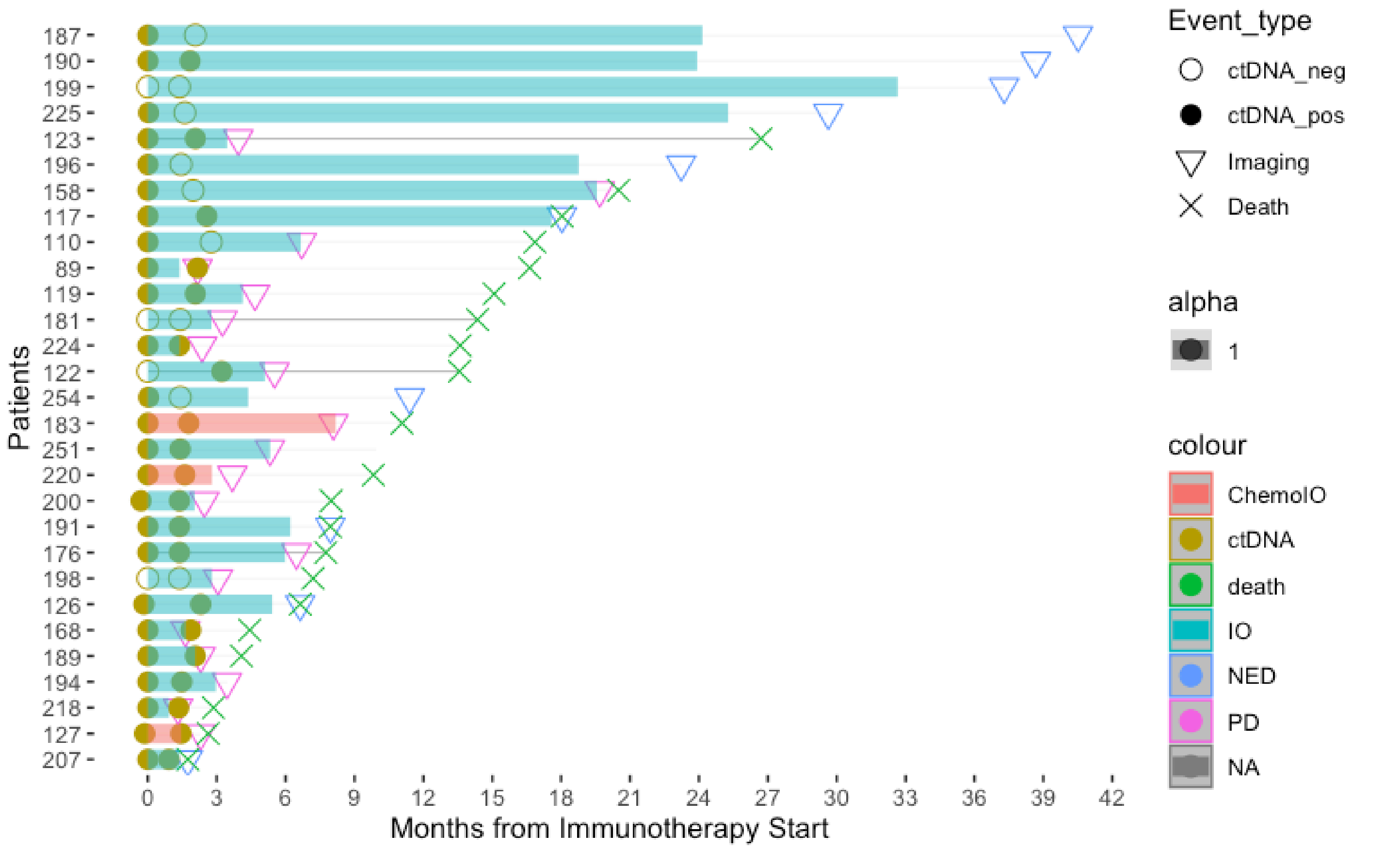
```
setwd("~/Downloads")
clinstage <- read.csv("EORTC ICI_OP.csv")
clinstage_df <- as.data.frame(clinstage)

# Creating the basic swimmer plot
oplot <- swimmer_plot(df=clinstage_df,
                      id='PatientName',
                      end='fu.diff.months',
                      fill='gray',
                      width=.01)

# Adding themes and scales
oplot <- oplot + theme(panel.border = element_blank())
oplot <- oplot + scale_y_continuous(breaks = seq(0, 48, by = 3))
oplot <- oplot + labs(x ="Patients", y="Months from Immunotherapy Start")

# Adding swimmer points
oplot_ev1 <- oplot + swimmer_points(df_points=clinstage_df,
                                    id='PatientName',
                                    time='date.diff.months',
                                    name_shape ='Event_type',
                                    name_col = 'Event',
                                    size=3.5,fill='black')
# Optionally uncomment and use col='darkgreen' if needed

# Adding shape manual scale
oplot_ev1.1 <- oplot_ev1 + ggplot2::scale_shape_manual(name="Event_type",
                                                       values=c(1,16,6,4),
                                                       breaks=c('ctDNA_neg','ctDNA_po
s','Imaging','Death'))

# Display the plot
oplot_ev1.1
```



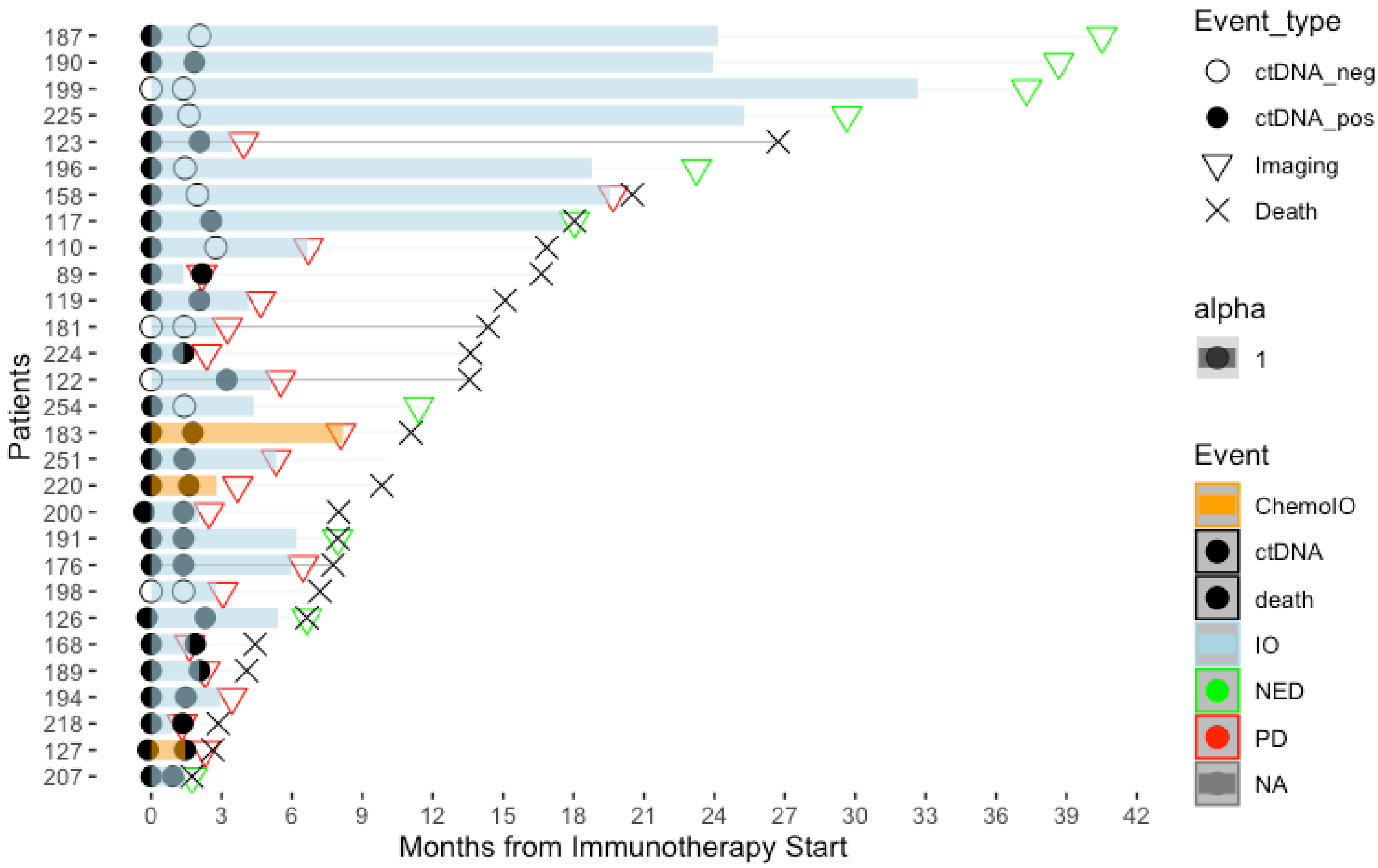
Hide

```
                                        oplot_ev2 <- oplot_ev1.1 + swimmer_lines(df_lines=cl
instage_df,
                                                                                        id='Patient
Name',
                                                                                        start='Tx_s
tart.months',
                                                                                        end='Tx_en
d.months',
                                                                                        name_col='T
x_type',
                                                                                        size=3.5,
                                                                                        name_alpha
= 1.0)
                                        oplot_ev2 <- oplot_ev2 + guides(linetype = guide_leg
end(override.aes = list(size = 5, color = "black")))
                                        oplot_ev2
```



Hide

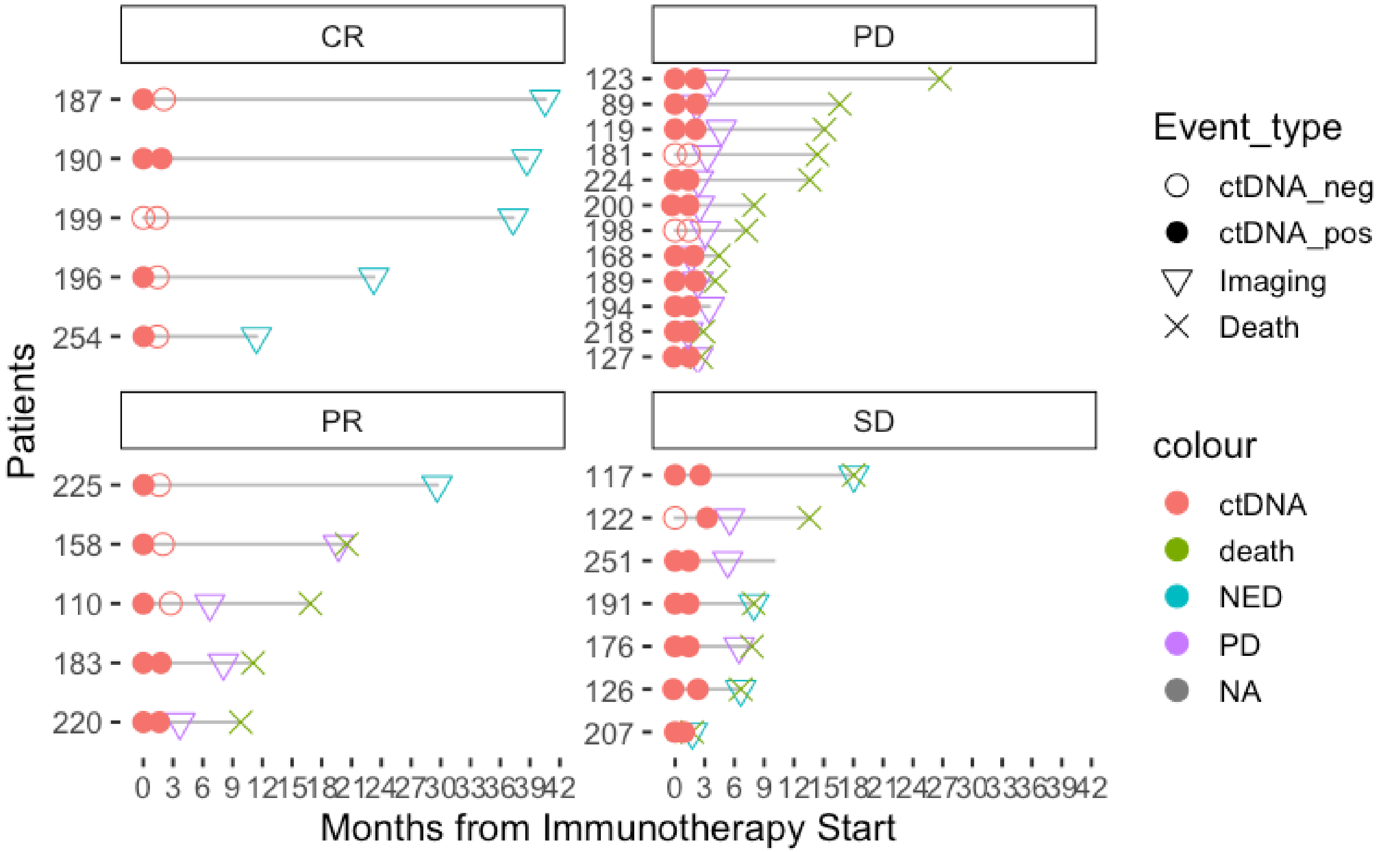
```
oplot_ev2.2 <- oplot_ev2 + ggplot2::scale_color_manual(name="Event",values=c( "orange",
"black", "black", "lightblue", "green", "red"))
oplot_ev2.2
```



#Overview plot - stratified by BOR

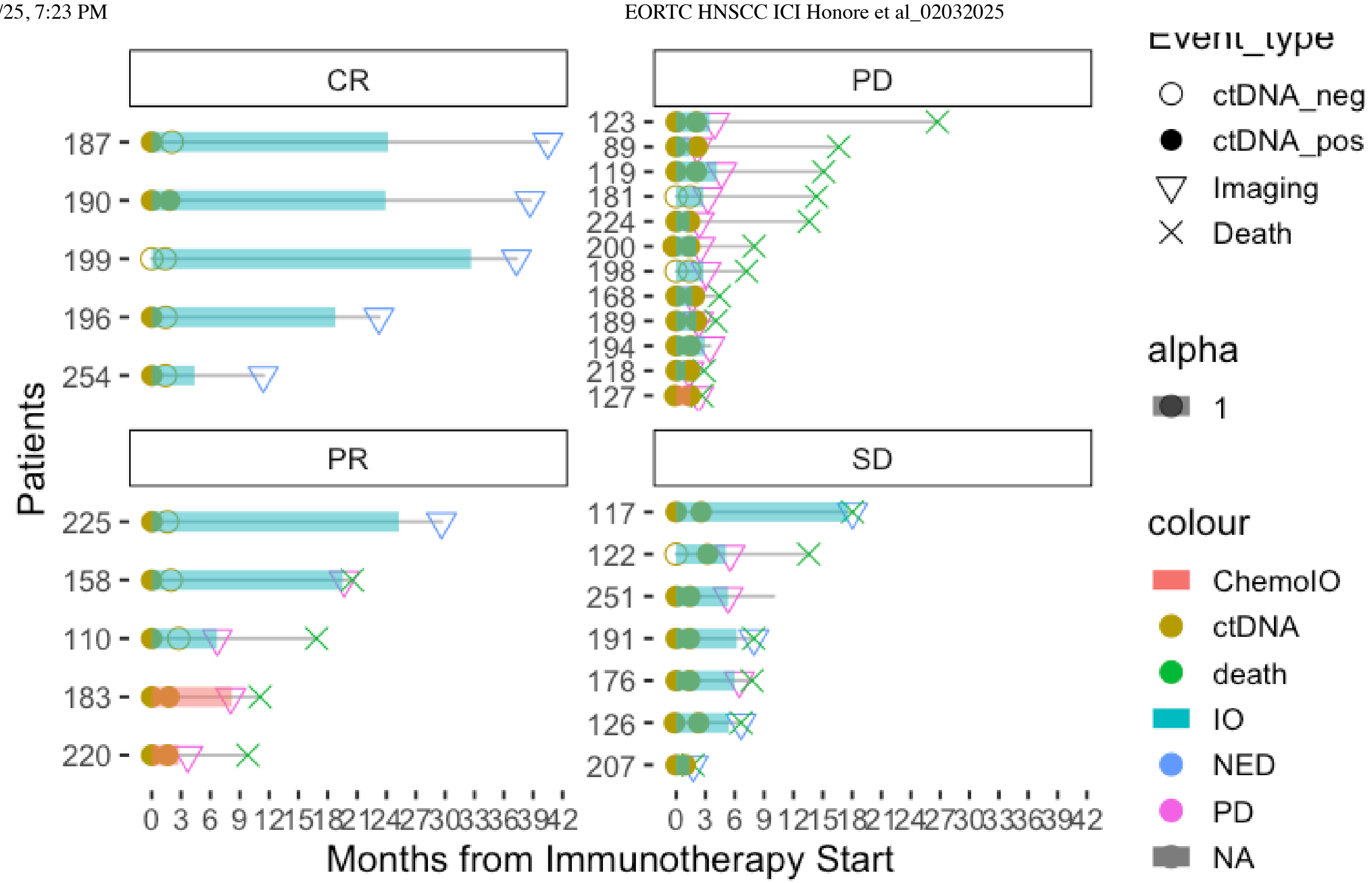
Hide

```
setwd("~/Downloads")
clinstage <- read.csv("EORTC ICI_OP.csv")
clinstage_df <- as.data.frame(clinstage)

# Creating the basic swimmer plot
oplot_stratify <-swimmer_plot(df=clinstage_df,
                              id='PatientName',
                              end='fu.diff.months',
                              col="gray",
                              alpha=0.75,
                              width=.01,
                              base_size = 14,
                              stratify= c('RECIST'))
oplot_stratify <- oplot_stratify + theme(panel.border = element_blank())
oplot_stratify <- oplot_stratify + scale_y_continuous(breaks = seq(0, 42, by = 3))
oplot_stratify <- oplot_stratify + labs(x ="Patients" , y="Months from Immunotherapy Sta
rt")

# Adding swimmer points
oplot_ev1 <- oplot_stratify + swimmer_points(df_points=clinstage_df,
                                    id='PatientName',
                                    time='date.diff.months',
                                    name_shape ='Event_type',
                                    name_col = 'Event',
                                    size=3.5,fill='black')
# Optionally uncomment and use col='darkgreen' if needed

# Adding shape manual scale
oplot_ev1.1 <- oplot_ev1 + ggplot2::scale_shape_manual(name="Event_type",
                                                       values=c(1,16,6,4),
                                                       breaks=c('ctDNA_neg','ctDNA_po
s','Imaging','Death'))

# Display the plot
oplot_ev1.1
```



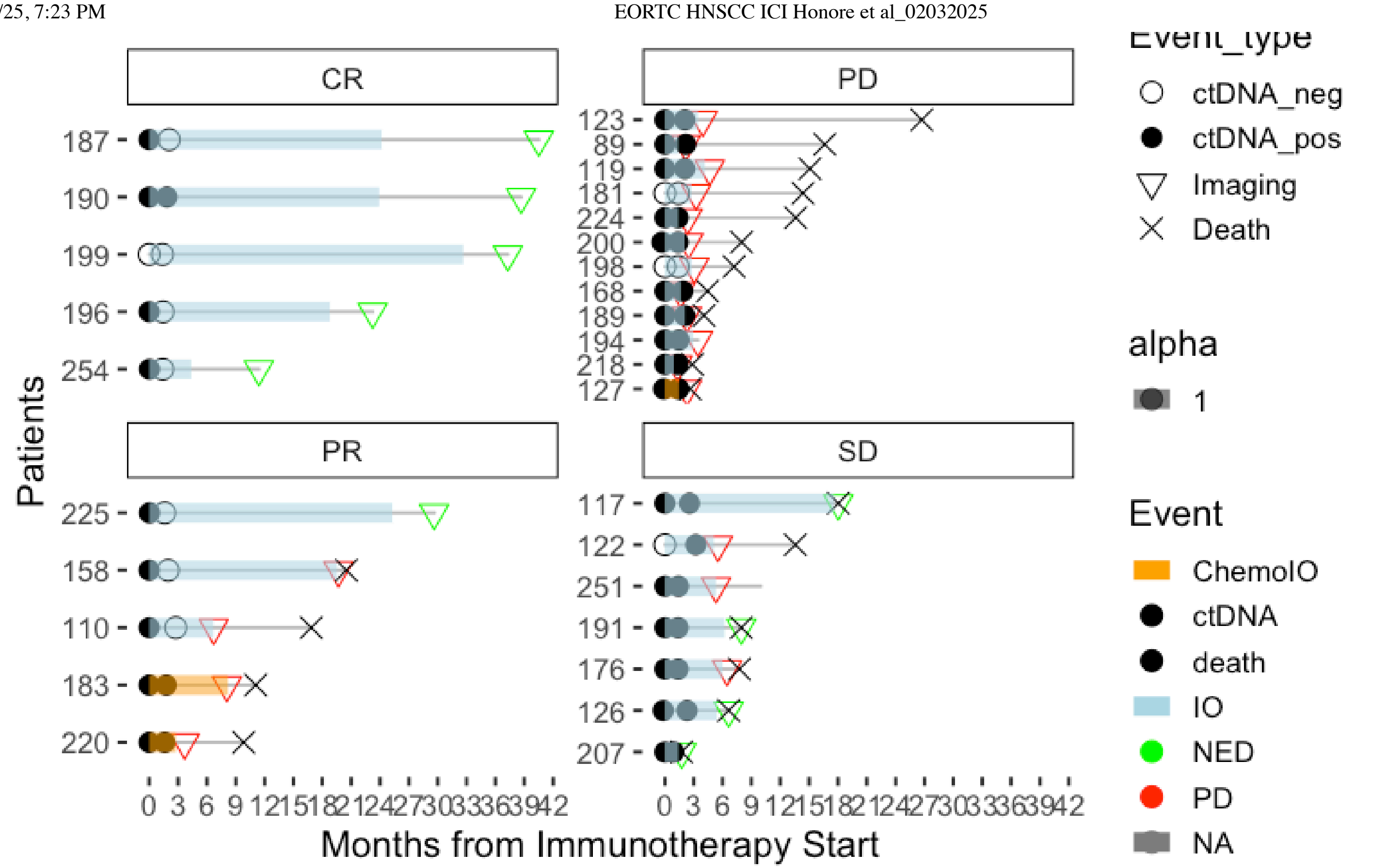
Hide

```
oplot_ev2 <- oplot_ev1.1 + swimmer_lines(df_lines=clinstage_df,
                                         id='Patient Name',
                                         start='Tx_start.months',
                                         end='Tx_end.months',
                                         name_col='Tx_type',
                                         size=3.5,
                                         name_alpha = 1.0)
oplot_ev2 <- oplot_ev2 + guides(linetype = guide_legend(override.aes = list(size = 5, color = "black")))
oplot_ev2
```



Hide

```
oplot_ev2.2 <- oplot_ev2 + ggplot2::scale_color_manual(name="Event",values=c( "orange",
"black", "black", "lightblue", "green", "red"))
oplot_ev2.2
```



#Association of Baseline ctDNA MTM levels with clinicopathological factors

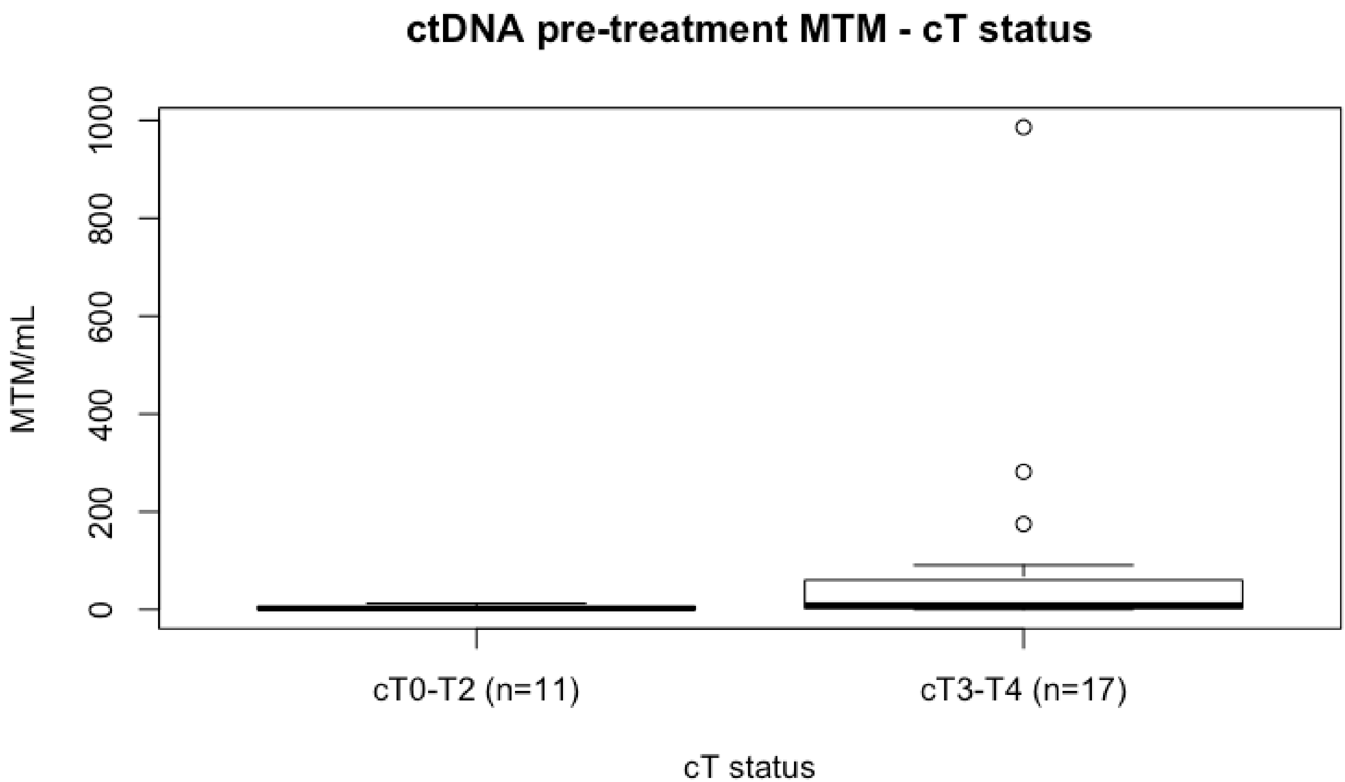
Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_datadf <- as.data.frame(circ_data)

tally(~cT.Status, data=circ_data, margins = TRUE)
```

```
cT.Status
cT0-T2 cT3-T4    cTx  Total
    11     17      1     29
```

Hide

```
circ_data$cT.Status <- factor(circ_data$cT.Status, levels = c("cT0-T2","cT3-T4"), labels
= c("cT0-T2 (n=11)","cT3-T4 (n=17)"))
boxplot(ctDNA.Base.MTM~cT.Status, data=circ_data, main="ctDNA pre-treatment MTM - cT sta
tus", xlab="cT status", ylab="MTM/mL", col="white",border="black")
```



Hide

```
m3<-wilcox.test(ctDNA.Base.MTM ~ cT.Status, data=circ_data, na.rm=TRUE, exact=FALSE, con
f.int=TRUE)
print(m3)
```

```

    Wilcoxon rank sum test with continuity correction

data:  ctDNA.Base.MTM by cT.Status
W = 45.5, p-value = 0.02523
alternative hypothesis: true location shift is not equal to 0
95 percent confidence interval:
 -57.0000356  -0.7999756
sample estimates:
difference in location
             -7.400021
```

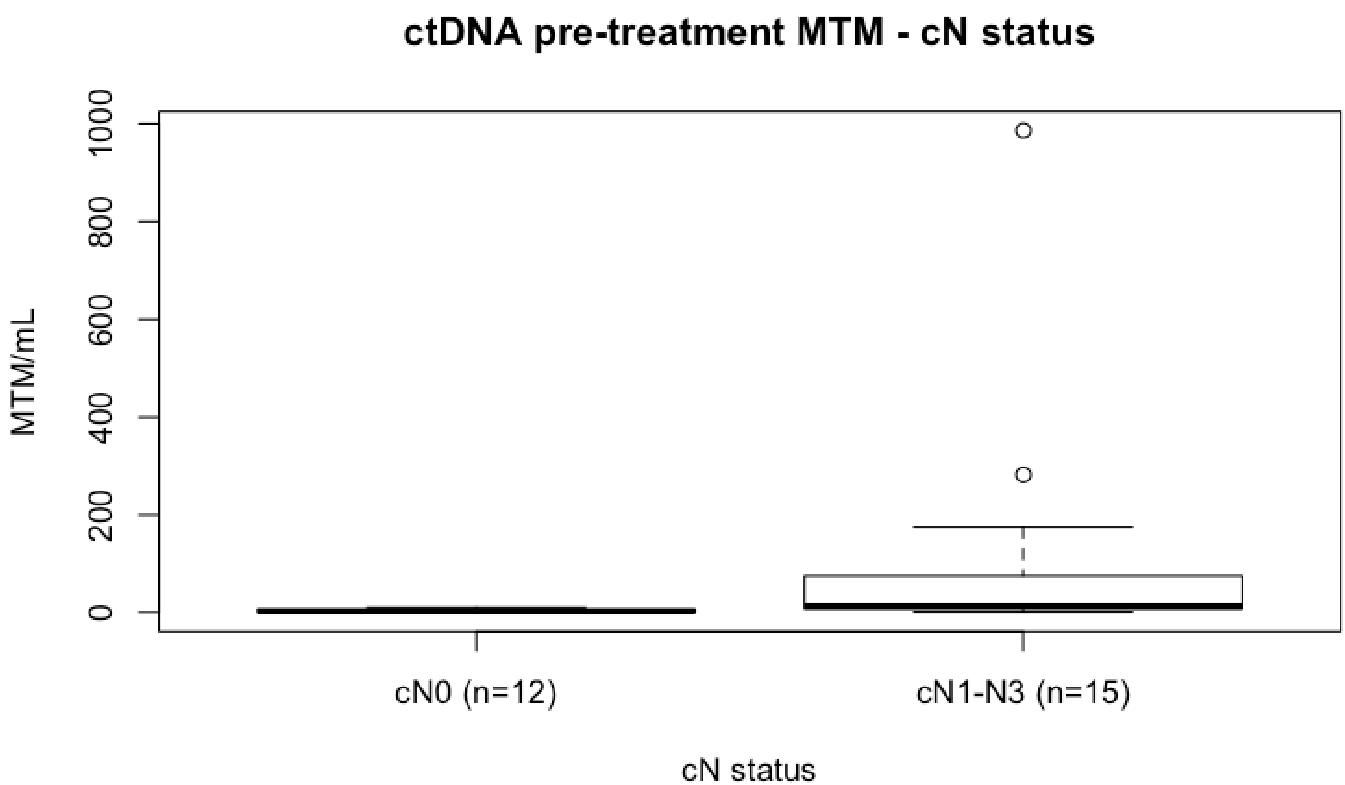
Hide

```
tally(~cN.Status, data=circ_data, margins = TRUE)
```

```
cN.Status
   cN0 cN1-N3     cNx  Total
    12     15       2     29
```

Hide

```
circ_data$cN.Status <- factor(circ_data$cN.Status, levels = c("cN0","cN1-N3"), labels =
c("cN0 (n=12)","cN1-N3 (n=15)"))
boxplot(ctDNA.Base.MTM~cN.Status, data=circ_data, main="ctDNA pre-treatment MTM - cN sta
tus", xlab="cN status", ylab="MTM/mL", col="white",border="black")
```



Hide

```
m4<-wilcox.test(ctDNA.Base.MTM ~ cN.Status, data=circ_data, na.rm=TRUE, exact=FALSE, con
f.int=TRUE)
print(m4)
```

```

	Wilcoxon rank sum test with continuity correction

data:  ctDNA.Base.MTM by cN.Status
W = 32.5, p-value = 0.005379
alternative hypothesis: true location shift is not equal to 0
95 percent confidence interval:
 -60.200006  -3.299997
sample estimates:
difference in location 
             -10.15193 
```

#Median MTM/mL levels for ctDNA positive pts pre-treatment

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data <- circ_data[circ_data$ctDNA.Base=="POSITIVE",]

median_ctDNA <- median(circ_data$ctDNA.Base.MTM, na.rm = TRUE)
range_ctDNA <- range(circ_data$ctDNA.Base.MTM, na.rm = TRUE)
cat("Median MTM/mL post-treatment:", median_ctDNA, "\n")
```

```
Median MTM/mL post-treatment: 8.4
```

Hide

```
cat("Range MTM/mL post-treatment:", range_ctDNA[1], "-", range_ctDNA[2], "\n")
```

```
Range MTM/mL post-treatment: 0.2 - 986
```

#Median MTM/mL levels for ctDNA positive pts post-treatment

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.postTx!="",]
circ_data <- circ_data[circ_data$ctDNA.postTx=="POSITIVE",]

median_ctDNA <- median(circ_data$ctDNA.postTx.MTM, na.rm = TRUE)
range_ctDNA <- range(circ_data$ctDNA.postTx.MTM, na.rm = TRUE)
cat("Median MTM/mL post-treatment:", median_ctDNA, "\n")
```

```
Median MTM/mL post-treatment: 7.9
```

Hide

```
cat("Range MTM/mL post-treatment:", range_ctDNA[1], "-", range_ctDNA[2], "\n")
```

```
Range MTM/mL post-treatment: 0.1 - 737.8
```

#Median time from end treatment to progression for ctDNA negative pts post-treatment

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.postTx!="",]
circ_data <- circ_data[circ_data$ctDNA.postTx=="NEGATIVE",]
circ_data <- circ_data[circ_data$PFS.Event=="TRUE",]

median_PFS <- median(circ_data$PFS.months, na.rm = TRUE)
range_PFS <- range(circ_data$PFS.months, na.rm = TRUE)
cat("Median PFS:", median_PFS, "\n")
```

```
Median PFS: 4.977494
```

Hide

```
cat("Range PFS:", range_PFS[1], "-", range_PFS[2], "\n")
```

```
Range PFS: 3.055492 - 19.67999
```

#PFS by ctDNA status post/during-ICI

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.postTx!="",]
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)~ctDNA.postTx, dat
a = circ_data)
```

```
Call: survfit(formula = Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event) ~ 
    ctDNA.postTx, data = circ_data)

                       n events median 0.95LCL 0.95UCL
ctDNA.postTx=NEGATIVE  9      4     NA    6.70      NA
ctDNA.postTx=POSITIVE 20     19   3.81    2.37    6.64
```

Hide

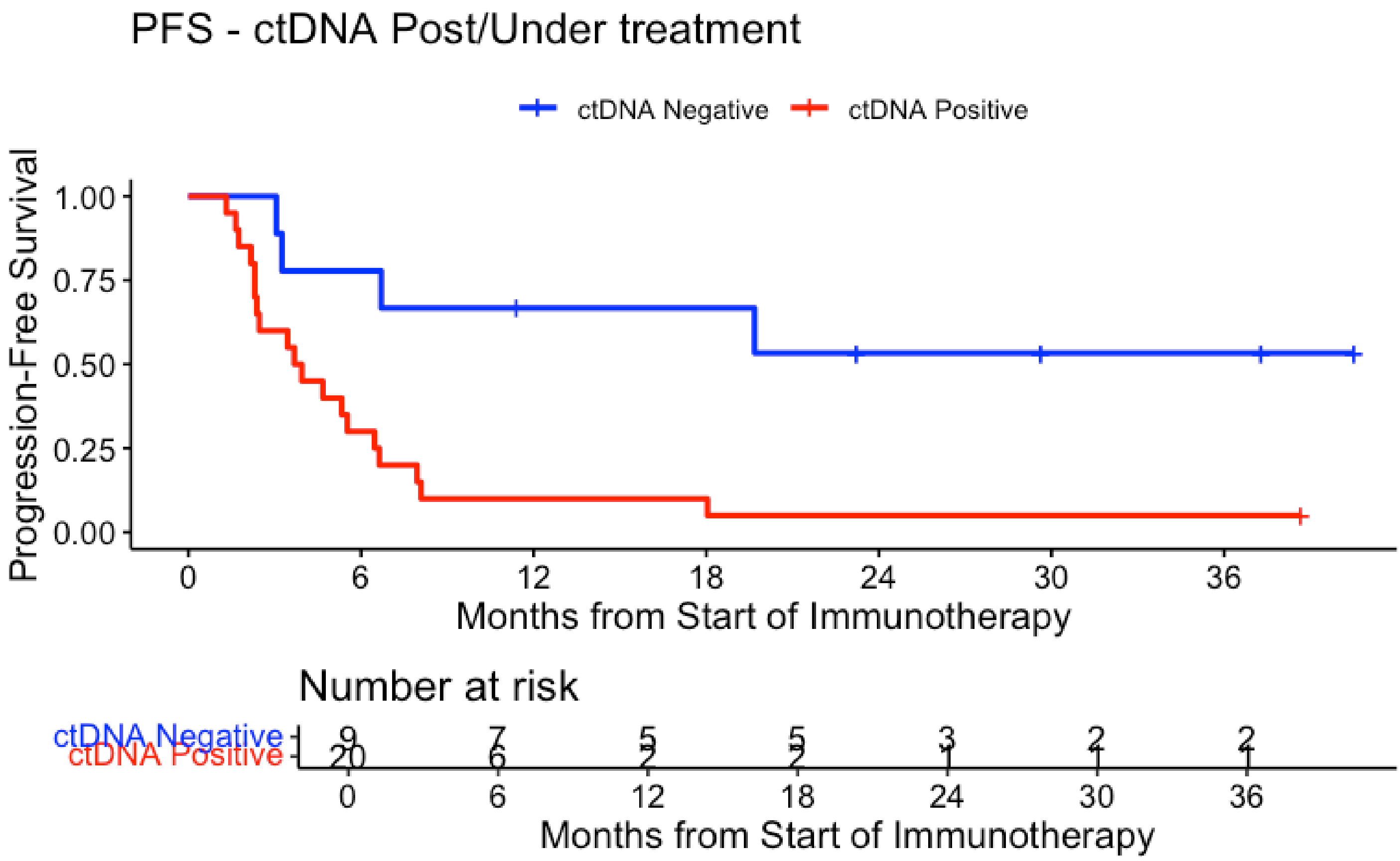
```
event_summary <- circ_data %>%
  group_by(ctDNA.postTx) %>%
  summarise(
    Total = n(),
    Events = sum(PFS.Event),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

ctDNA.postTx	Total	Events	Fraction	Percentage
<chr>	<int>	<int>	<dbl>	<dbl>
NEGATIVE	9	4	0.4444444	44.44444
POSITIVE	20	19	0.9500000	95.00000

2 rows

Hide

```
surv_object <-Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.postTx, data = circ_data,conf.int=0.95,conf.type
="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRU
E, break.time.by=6, palette=c("blue","red"), title="PFS - ctDNA Post/Under treatment", y
lab= "Progression-Free Survival", xlab="Months from Start of Immunotherapy", legend.labs
=c("ctDNA Negative", "ctDNA Positive"), legend.title="")
```



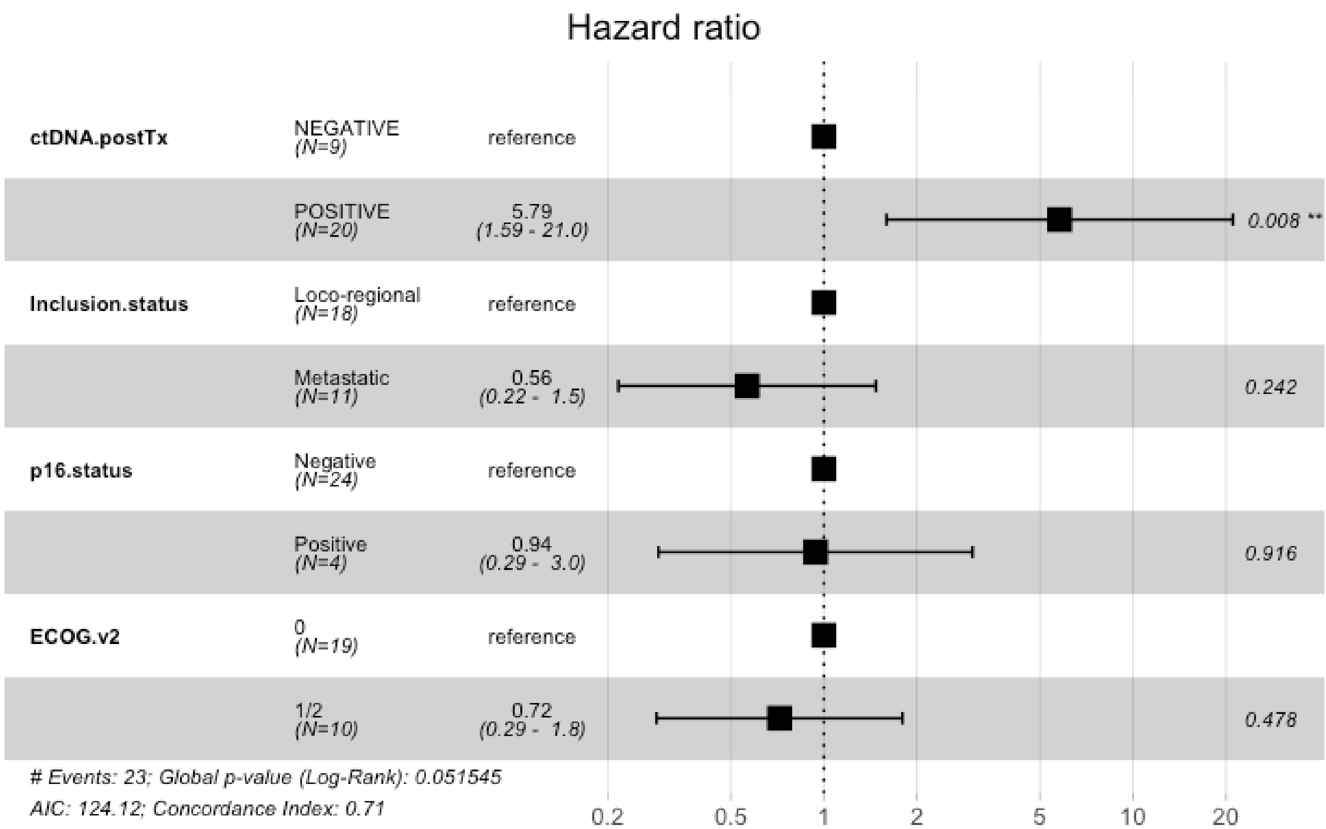
Hide

```
summary(KM_curve, times= c(0, 12, 24, 36))
```

```
Call: survfit(formula = surv_object ~ ctDNA.postTx, data = circ_data,
    conf.int = 0.95, conf.type = "log-log")

                ctDNA.postTx=NEGATIVE
 time n.risk n.event survival std.err lower 95% CI upper 95% CI
    0      9       0    1.000   0.000        1.000        1.000
   12      5       3    0.667   0.157        0.282        0.878
   24      3       1    0.533   0.173        0.177        0.796
   36      2       0    0.533   0.173        0.177        0.796

                ctDNA.postTx=POSITIVE
 time n.risk n.event survival std.err lower 95% CI upper 95% CI
    0     20       0     1.00  0.0000      1.00000        1.000
   12      2      18     0.10  0.0671      0.01698        0.272
   24      1       1     0.05  0.0487      0.00345        0.205
   36      1       0     0.05  0.0487      0.00345        0.205
```

Hide

```
circ_data$ctDNA.postTx <- factor(circ_data$ctDNA.postTx, levels=c("NEGATIVE","POSITIV
E"))
circ_data$Inclusion.status <- factor(circ_data$Inclusion.status, levels = c("Loco-region
al", "Metastatic"))
circ_data$p16.status <- factor(circ_data$p16.status, levels = c("Negative", "Positive"))
circ_data$ECOG.v2 <- factor(circ_data$ECOG.v2, levels = c("0", "1/2"))
cox_fit <- coxph(surv_object ~ ctDNA.postTx + Inclusion.status + p16.status + ECOG.v2, d
ata=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ ctDNA.postTx + Inclusion.status +
    p16.status + ECOG.v2, data = circ_data)

  n= 28, number of events= 23
   (1 observation deleted due to missingness)

                              coef exp(coef) se(coef)      z Pr(>|z|)
ctDNA.postTxPOSITIVE        1.7556    5.7870   0.6584  2.666  0.00767 **
Inclusion.statusMetastatic -0.5723    0.5642   0.4895 -1.169  0.24237
p16.statusPositive         -0.0633    0.9387   0.5971 -0.106  0.91557
ECOG.v21/2                 -0.3318    0.7176   0.4674 -0.710  0.47770
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

                           exp(coef) exp(-coef) lower .95 upper .95
ctDNA.postTxPOSITIVE          5.7870     0.1728    1.5922    21.033
Inclusion.statusMetastatic    0.5642     1.7724    0.2162     1.473
p16.statusPositive            0.9387     1.0653    0.2912     3.025
ECOG.v21/2                    0.7176     1.3935    0.2871     1.794

Concordance= 0.708  (se = 0.053 )
Likelihood ratio test= 9.41  on 4 df,   p=0.05
Wald test            = 7.4  on 4 df,   p=0.1
Score (logrank) test = 8.26  on 4 df,   p=0.08
```

Hide

```
cox_fit_summary <- summary(cox_fit)

circ_data$ctDNA.postTx <- factor(circ_data$ctDNA.postTx, levels = c("NEGATIVE", "POSITIV
E"), labels = c("Negative", "Positive"))
circ_data$PFS.Event <- factor(circ_data$PFS.Event, levels = c("FALSE", "TRUE"), labels =
c("No Progression", "Progression"))
contingency_table <- table(circ_data$ctDNA.postTx, circ_data$PFS.Event)
chi_square_test <- chisq.test(contingency_table)
```

```
 G2; H2;Warning h in stats::chisq.test(x, y, ...) :
  Chi-squared approximation may be incorrect g
```

Hide

```
print(chi_square_test)
```

```
	Pearson's Chi-squared test with Yates' continuity correction

data:  contingency_table
X-squared = 6.8323, df = 1, p-value = 0.008952
```

Hide

```
fisher_exact_test <- fisher.test(contingency_table)
print(fisher_exact_test)
```

```

	Fisher's Exact Test for Count Data

data:  contingency_table
p-value = 0.005482
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
    1.68957 1169.09723
sample estimates:
odds ratio 
  20.33413 
```

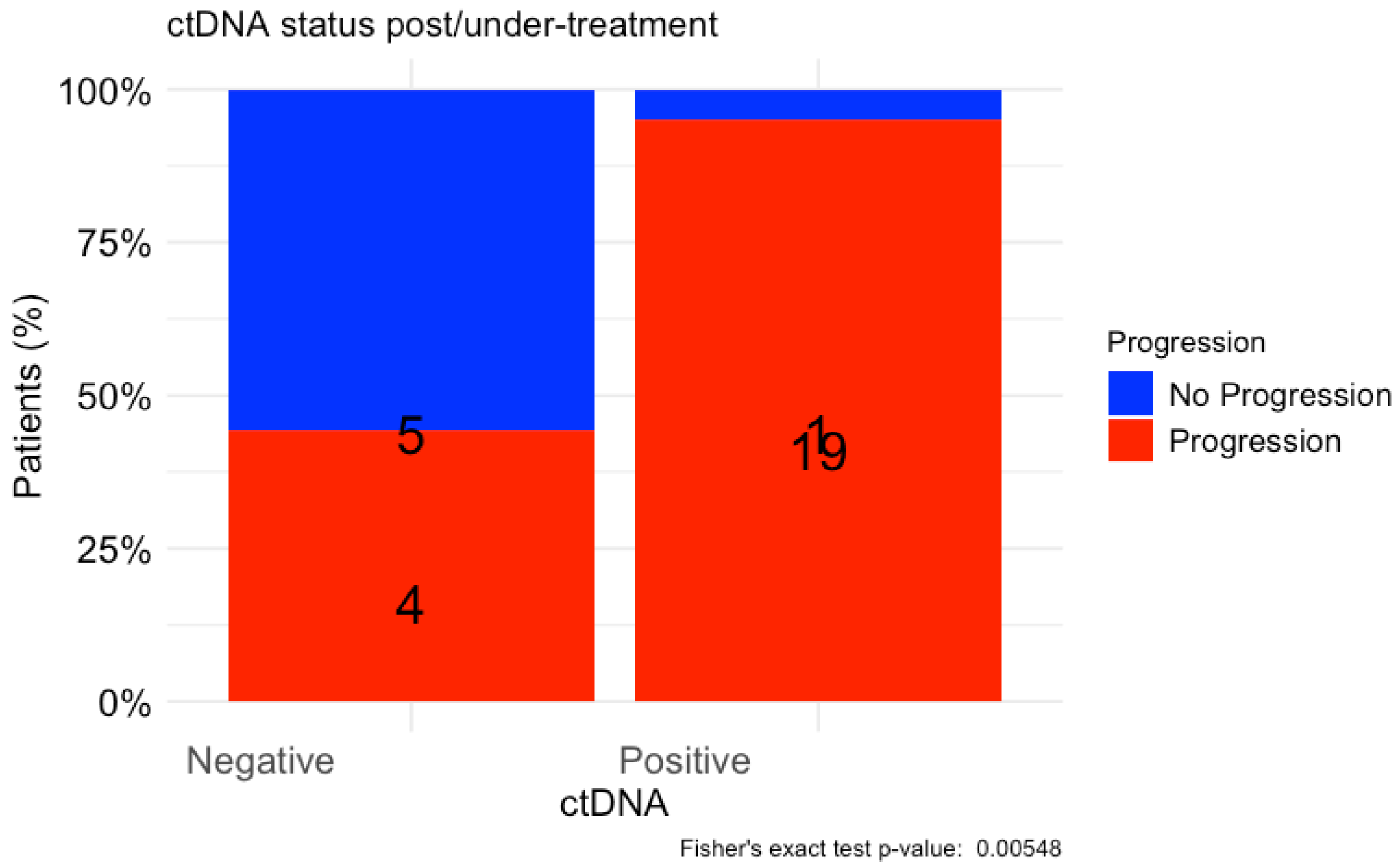
Hide

```
print(contingency_table)
```

```

          No Progression Progression
  Negative              5           4
  Positive              1          19
```

Hide

```
table_df <- as.data.frame(contingency_table)
table_df$Total <- ave(table_df$Freq, table_df$Var1, FUN = sum)
table_df$Percentage <- table_df$Freq / table_df$Total
table_df$MiddlePercentage <- table_df$Percentage / 2
ggplot(table_df, aes(x = Var1, y = Percentage, fill = Var2)) +
  geom_bar(stat = "identity") +
  geom_text(aes(y = MiddlePercentage, label = Freq), position = "stack", color = "blac
k", vjust = 1.5, size = 7) +
  theme_minimal() +
  labs(title = "ctDNA status post/under-treatment",
       x = "ctDNA",
       y = "Patients (%)",
       fill = "Progression",
       caption = paste("Fisher's exact test p-value: ", format.pval(fisher_exact_test$p.
value))) +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_manual(values = c("No Progression" = "blue", "Progression" = "red")) + # de
fine custom colors
  theme(axis.text.x = element_text(angle = 0, hjust = 1.5, size = 14), # increase x-axis
text size
        axis.text.y = element_text(size = 14, color = "black"), # increase y-axis text s
ize
        axis.title.x = element_text(size = 14, color = "black"), # increase x-axis label
size
        axis.title.y = element_text(size = 14, color = "black"), # increase y-axis label
size
        legend.text = element_text(size = 12, color = "black"))  # increase Progression
label size
```



#OS by ctDNA status post/during-ICI

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.postTx!="",]
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$OS.months, event = circ_data$OS.Event)~ctDNA.postTx, data
= circ_data)
```

```
Call: survfit(formula = Surv(time = circ_data$OS.months, event = circ_data$OS.Event) ~
    ctDNA.postTx, data = circ_data)

                       n events median 0.95LCL 0.95UCL
ctDNA.postTx=NEGATIVE  9      4     NA   16.85      NA
ctDNA.postTx=POSITIVE 20     17   10.4    7.75      18
```

Hide

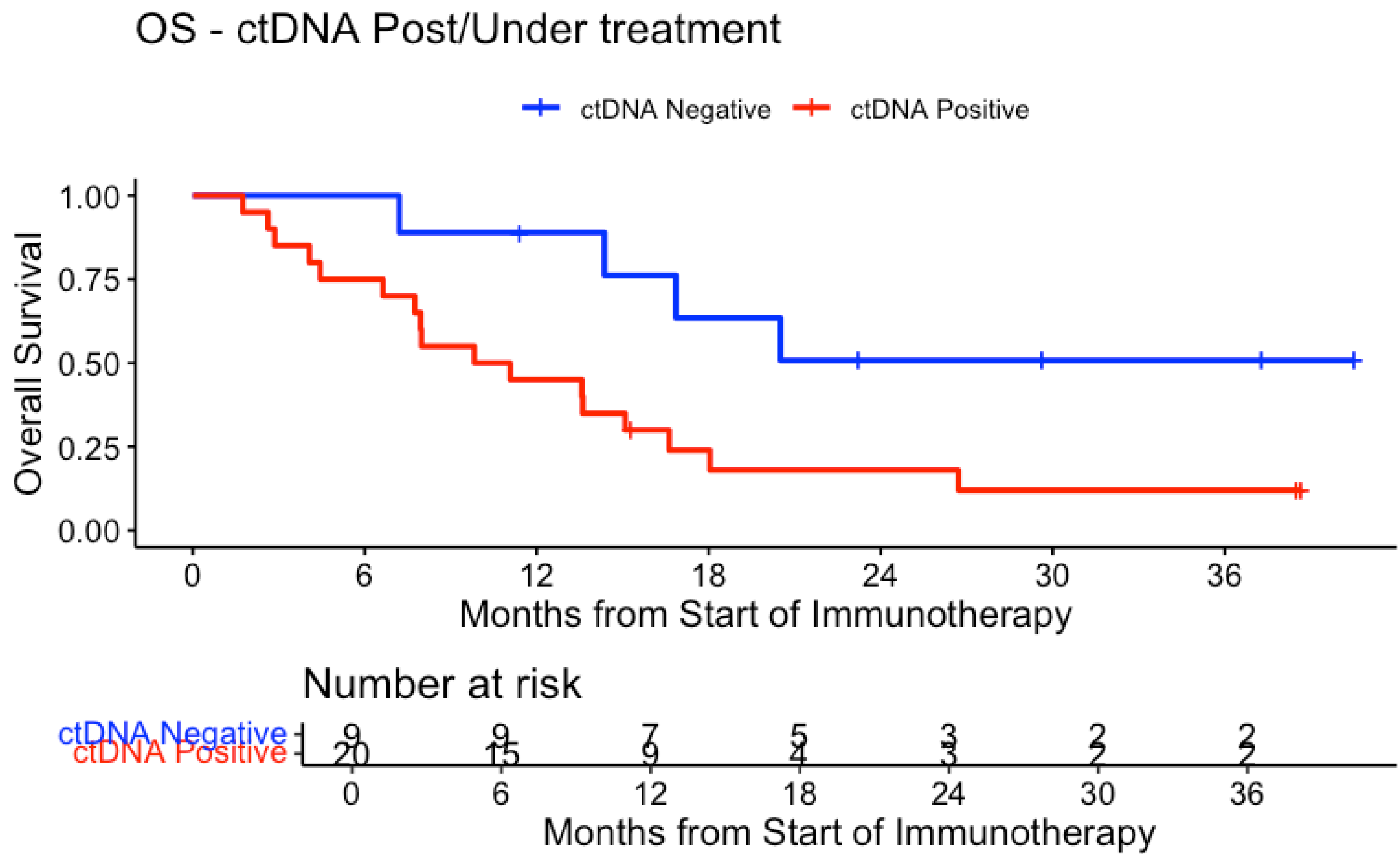
```
event_summary <- circ_data %>%
  group_by(ctDNA.postTx) %>%
  summarise(
    Total = n(),
    Events = sum(OS.Event),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

ctDNA.postTx <chr>	Total <int>	Events <int>	Fraction <dbl>	Percentage <dbl>
NEGATIVE	9	4	0.4444444	44.44444
POSITIVE	20	17	0.8500000	85.00000

2 rows

Hide

```
surv_object <-Surv(time = circ_data$OS.months, event = circ_data$OS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.postTx, data = circ_data,conf.int=0.95,conf.type
="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRU
E, break.time.by=6, palette=c("blue","red"), title="OS - ctDNA Post/Under treatment", yl
ab= "Overall Survival", xlab="Months from Start of Immunotherapy", legend.labs=c("ctDNA
Negative", "ctDNA Positive"), legend.title="")
```



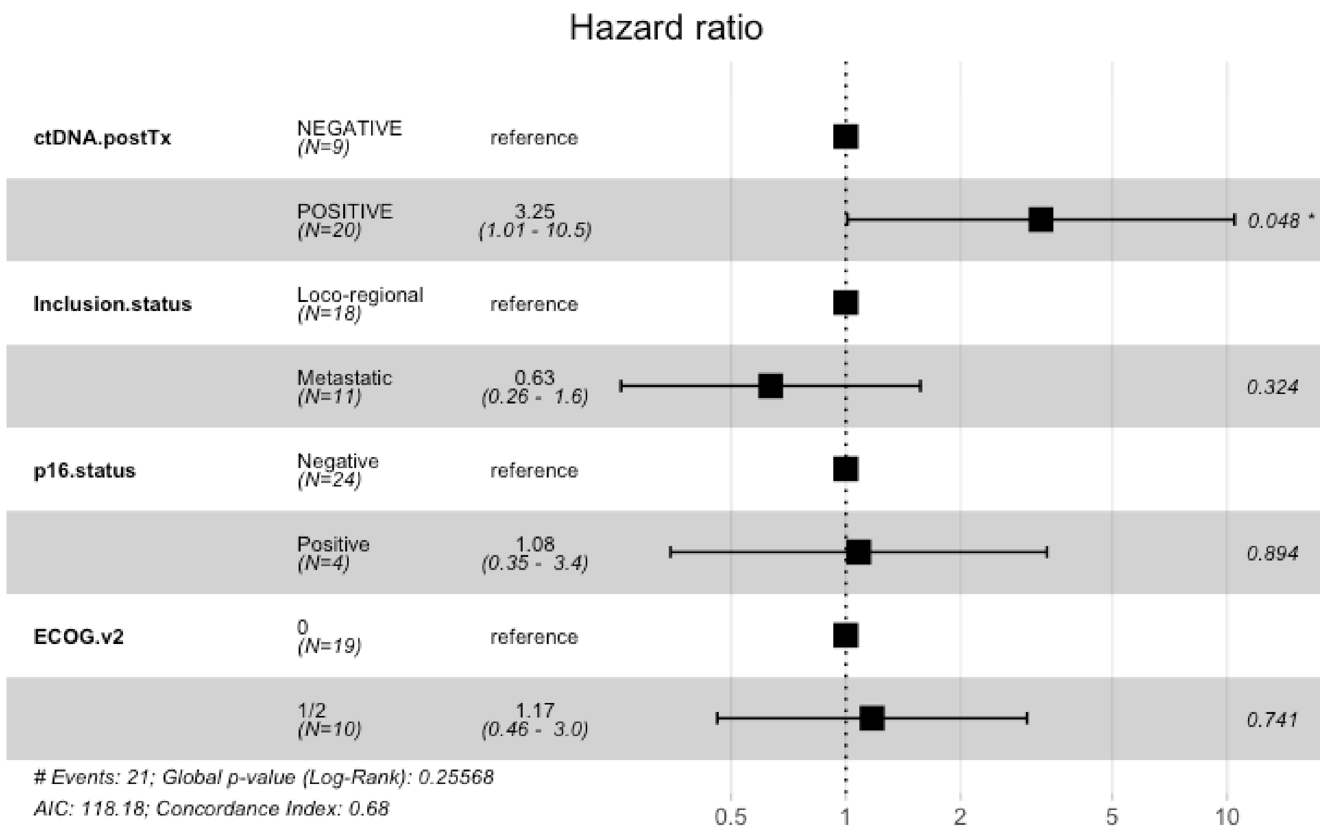
Hide

```
summary(KM_curve, times= c(12, 24, 36))
```

```
Call: survfit(formula = surv_object ~ ctDNA.postTx, data = circ_data,
    conf.int = 0.95, conf.type = "log-log")

                ctDNA.postTx=NEGATIVE
 time n.risk n.event survival std.err lower 95% CI upper 95% CI
   12      7       1    0.889   0.105        0.433        0.984
   24      3       3    0.508   0.177        0.157        0.781
   36      2       0    0.508   0.177        0.157        0.781

                ctDNA.postTx=POSITIVE
 time n.risk n.event survival std.err lower 95% CI upper 95% CI
   12      9      11     0.45  0.1112       0.2311        0.647
   24      3       5     0.18  0.0900       0.0480        0.380
   36      2       1     0.12  0.0775       0.0213        0.311
```

Hide

```
circ_data$ctDNA.postTx <- factor(circ_data$ctDNA.postTx, levels=c("NEGATIVE","POSITIV
E"))
circ_data$Inclusion.status <- factor(circ_data$Inclusion.status, levels = c("Loco-region
al", "Metastatic"))
circ_data$p16.status <- factor(circ_data$p16.status, levels = c("Negative", "Positive"))
circ_data$ECOG.v2 <- factor(circ_data$ECOG.v2, levels = c("0", "1/2"))
cox_fit <- coxph(surv_object ~ ctDNA.postTx + Inclusion.status + p16.status + ECOG.v2, d
ata=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ ctDNA.postTx + Inclusion.status +
    p16.status + ECOG.v2, data = circ_data)

  n= 28, number of events= 21
   (1 observation deleted due to missingness)

                               coef exp(coef) se(coef)      z Pr(>|z|)
ctDNA.postTxPOSITIVE        1.17775   3.24706  0.59654  1.974   0.0483 *
Inclusion.statusMetastatic -0.45503   0.63443  0.46155 -0.986   0.3242
p16.statusPositive          0.07725   1.08031  0.57990  0.133   0.8940
ECOG.v21/2                  0.15760   1.17070  0.47744  0.330   0.7413
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                           exp(coef) exp(-coef) lower .95 upper .95
ctDNA.postTxPOSITIVE          3.2471     0.3080    1.0086    10.454
Inclusion.statusMetastatic    0.6344     1.5762    0.2567     1.568
p16.statusPositive            1.0803     0.9257    0.3467     3.366
ECOG.v21/2                    1.1707     0.8542    0.4593     2.984

Concordance= 0.678  (se = 0.052 )
Likelihood ratio test= 5.32  on 4 df,   p=0.3
Wald test            = 4.59  on 4 df,   p=0.3
Score (logrank) test = 4.92  on 4 df,   p=0.3
```

Hide

```
cox_fit_summary <- summary(cox_fit)

circ_data$ctDNA.postTx <- factor(circ_data$ctDNA.postTx, levels = c("NEGATIVE", "POSITIV
E"), labels = c("Negative", "Positive"))
circ_data$OS.Event <- factor(circ_data$OS.Event, levels = c("FALSE", "TRUE"), labels = c
("Alive", "Deceased"))
contingency_table <- table(circ_data$ctDNA.postTx, circ_data$OS.Event)
chi_square_test <- chisq.test(contingency_table)
```

```
 G2; H2;Warning h in stats::chisq.test(x, y, ...) :
  Chi-squared approximation may be incorrect g
```

Hide

```
print(chi_square_test)
```

```


    Pearson's Chi-squared test with Yates' continuity correction

data:  contingency_table
X-squared = 3.2819, df = 1, p-value = 0.07005
```

Hide

```
fisher_exact_test <- fisher.test(contingency_table)
print(fisher_exact_test)
```

```

	Fisher's Exact Test for Count Data

data:  contingency_table
p-value = 0.0667
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
  0.8624077 62.5051555
sample estimates:
odds ratio
  6.505452
```

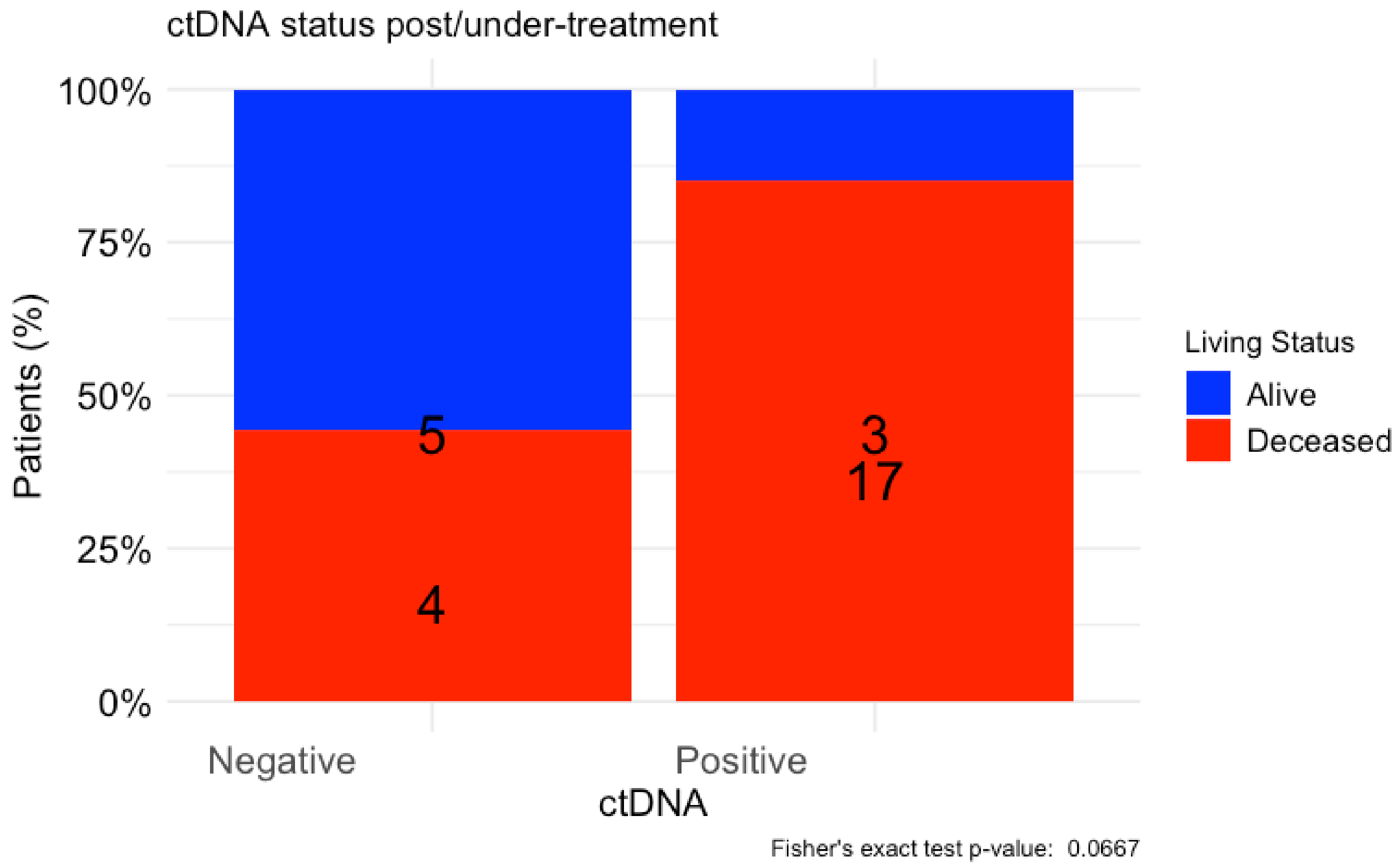
Hide

```
print(contingency_table)
```

```
           Alive Deceased
  Negative     5        4
  Positive     3       17
```

Hide

```
table_df <- as.data.frame(contingency_table)
table_df$Total <- ave(table_df$Freq, table_df$Var1, FUN = sum)
table_df$Percentage <- table_df$Freq / table_df$Total
table_df$MiddlePercentage <- table_df$Percentage / 2
ggplot(table_df, aes(x = Var1, y = Percentage, fill = Var2)) +
  geom_bar(stat = "identity") +
  geom_text(aes(y = MiddlePercentage, label = Freq), position = "stack", color = "blac
k", vjust = 1.5, size = 7) +
  theme_minimal() +
  labs(title = "ctDNA status post/under-treatment",
       x = "ctDNA",
       y = "Patients (%)",
       fill = "Living Status",
       caption = paste("Fisher's exact test p-value: ", format.pval(fisher_exact_test$p.
value))) +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_manual(values = c("Alive" = "blue", "Deceased" = "red")) + # define custom
colors
  theme(axis.text.x = element_text(angle = 0, hjust = 1.5, size = 14), # increase x-axis
text size
        axis.text.y = element_text(size = 14, color = "black"), # increase y-axis text s
ize
        axis.title.x = element_text(size = 14, color = "black"), # increase x-axis label
size
        axis.title.y = element_text(size = 14, color = "black"), # increase y-axis label
size
        legend.text = element_text(size = 12, color = "black"))  # increase Progression
label size
```



#Association of ctDNA status post/during-ICI with BOR

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_datadf <- as.data.frame(circ_data)

circ_data$ctDNA.postTx <- factor(circ_data$ctDNA.postTx, levels = c("NEGATIVE", "POSITIV
E"), labels = c("Negative", "Positive"))
circ_data$RESIST <- factor(circ_data$RESIST, levels = c("CR", "PR", "SD", "PD"))
contingency_table <- table(circ_data$ctDNA.postTx, circ_data$RESIST)
chi_square_test <- chisq.test(contingency_table)
```

```
G2; H2;Warning h in stats::chisq.test(x, y, ...) :
  Chi-squared approximation may be incorrect g
```

Hide

```
print(chi_square_test)
```

```
    Pearson's Chi-squared test

data:  contingency_table
X-squared = 11.869, df = 3, p-value = 0.007847
```

Hide

```
fisher_exact_test <- fisher.test(contingency_table)
print(fisher_exact_test)
```

```
    Fisher's Exact Test for Count Data

data:  contingency_table
p-value = 0.006541
alternative hypothesis: two.sided
```

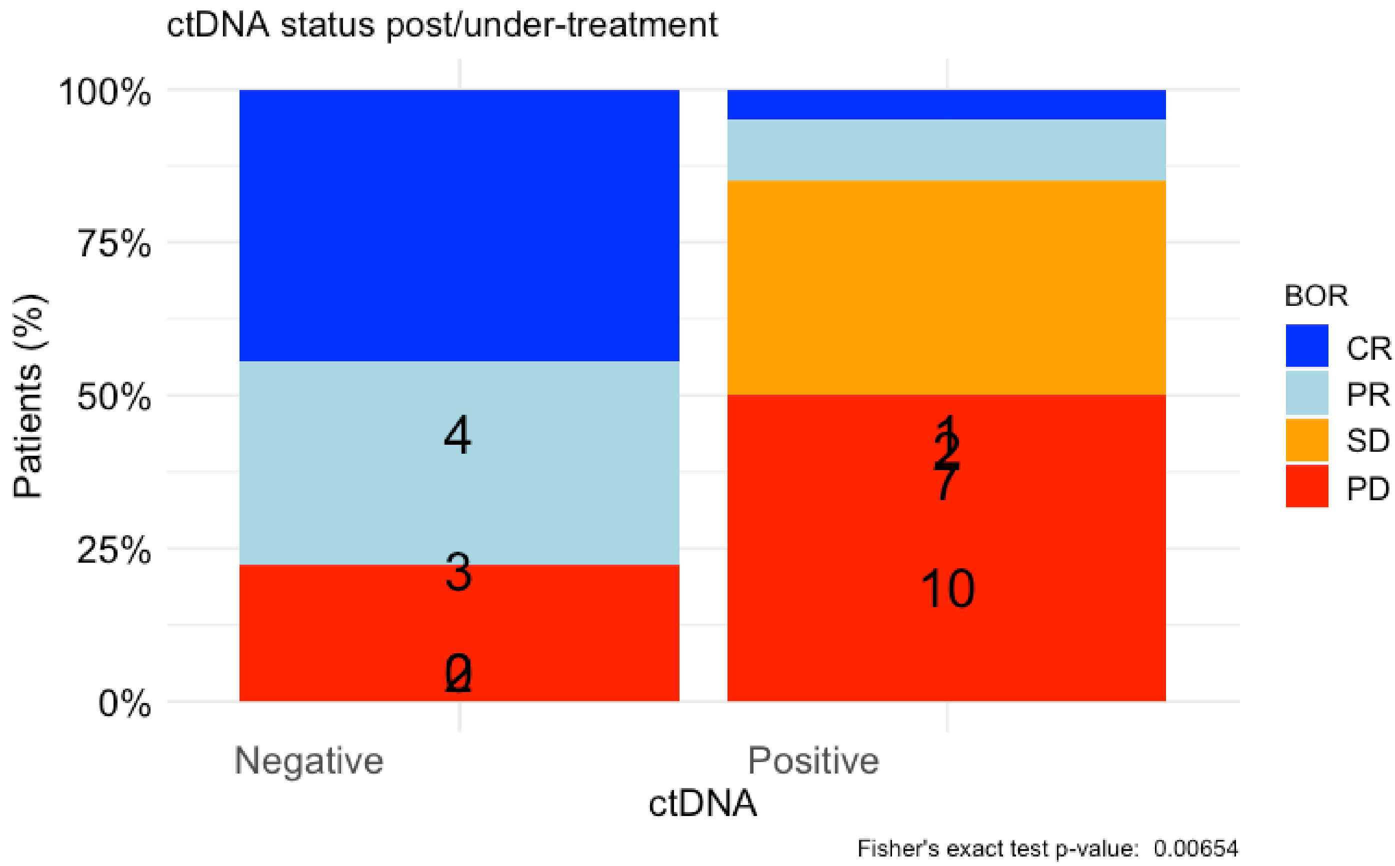
Hide

```
print(contingency_table)
```

```
           CR PR SD PD
  Negative  4  3  0  2
  Positive  1  2  7 10
```

Hide

```
table_df <- as.data.frame(contingency_table)
table_df$Total <- ave(table_df$Freq, table_df$Var1, FUN = sum)
table_df$Percentage <- table_df$Freq / table_df$Total
table_df$MiddlePercentage <- table_df$Percentage / 2
ggplot(table_df, aes(x = Var1, y = Percentage, fill = Var2)) +
  geom_bar(stat = "identity") +
  geom_text(aes(y = MiddlePercentage, label = Freq), position = "stack", color = "blac
k", vjust = 1.5, size = 7) +
  theme_minimal() +
  labs(title = "ctDNA status post/under-treatment",
       x = "ctDNA",
       y = "Patients (%)",
       fill = "BOR",
       caption = paste("Fisher's exact test p-value: ", format.pval(fisher_exact_test$p.
value))) +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_manual(values = c("CR" = "blue", "PR" = "lightblue", "SD" = "orange", "PD"
= "red")) + # define custom colors
  theme(axis.text.x = element_text(angle = 0, hjust = 1.5, size = 14), # increase x-axis
text size
        axis.text.y = element_text(size = 14, color = "black"), # increase y-axis text s
ize
        axis.title.x = element_text(size = 14, color = "black"), # increase x-axis label
size
        axis.title.y = element_text(size = 14, color = "black"), # increase y-axis label
size
        legend.text = element_text(size = 12, color = "black"))  # increase Progression
label size
```



#PFS by ctDNA kinetics post/during-ICI

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ΔctDNA!="",]
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)~ΔctDNA, data = ci
rc_data)
```

```
Call: survfit(formula = Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event) ~
    ΔctDNA, data = circ_data)

                  n events median 0.95LCL 0.95UCL
ΔctDNA=NEGATIVE 13      8  18.04     6.7      NA
ΔctDNA=POSITIVE 12     12   2.41     2.3      NA
```

Hide

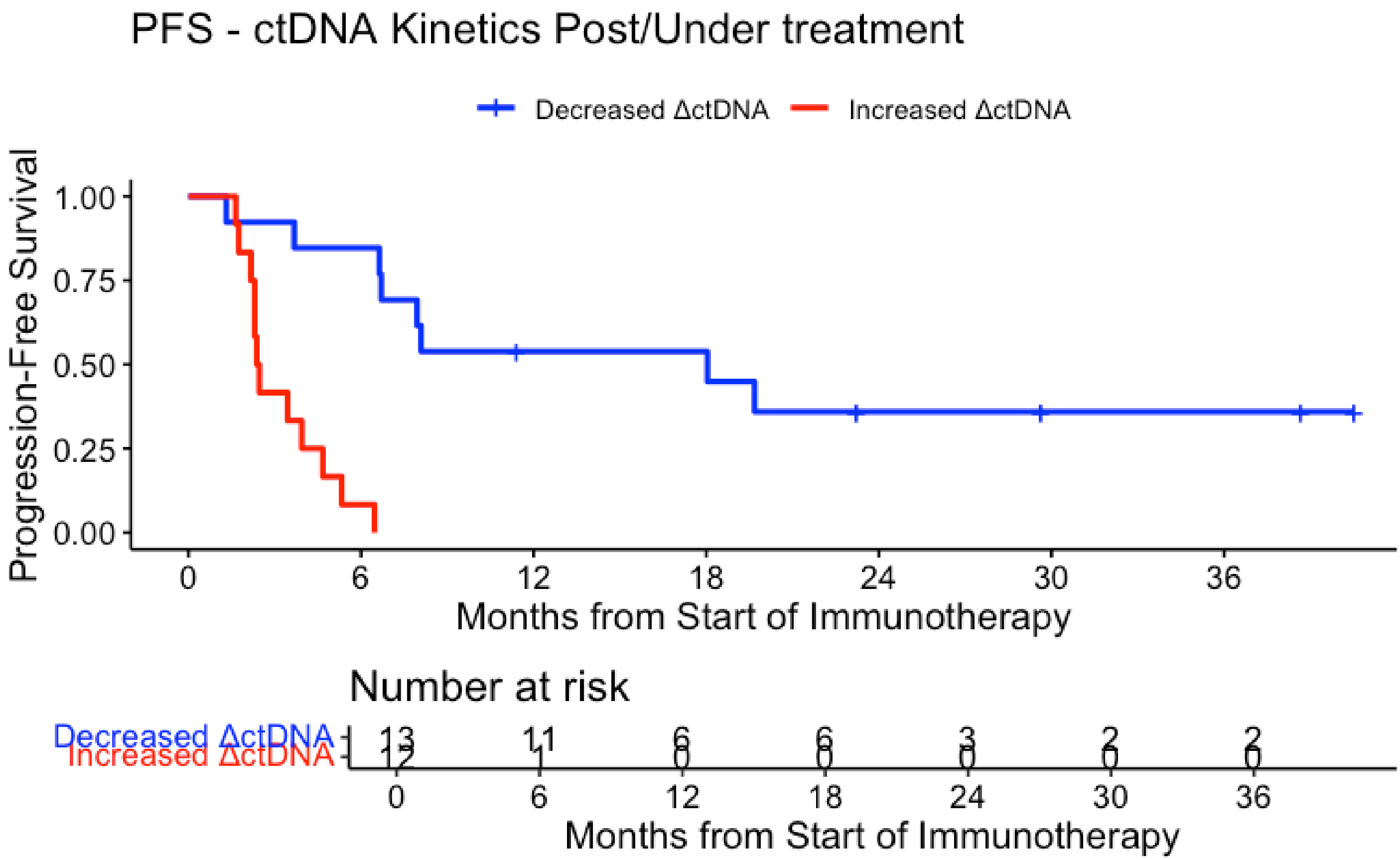
```
event_summary <- circ_data %>%
  group_by(ΔctDNA) %>%
  summarise(
    Total = n(),
    Events = sum(PFS.Event),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

ΔctDNA <chr>	Total <int>	Events <int>	Fraction <dbl>	Percentage <dbl>
NEGATIVE	13	8	0.6153846	61.53846
POSITIVE	12	12	1.0000000	100.00000

2 rows

Hide

```
surv_object <-Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
KM_curve <- survfit(surv_object ~ ΔctDNA, data = circ_data,conf.int=0.95,conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("blue","red"), title="PFS - ctDNA Kinetics Post/Under treatment", ylab= "Progression-Free Survival", xlab="Months from Start of Immunotherapy", legend.labs=c("Decreased ΔctDNA", "Increased ΔctDNA"), legend.title="")
```



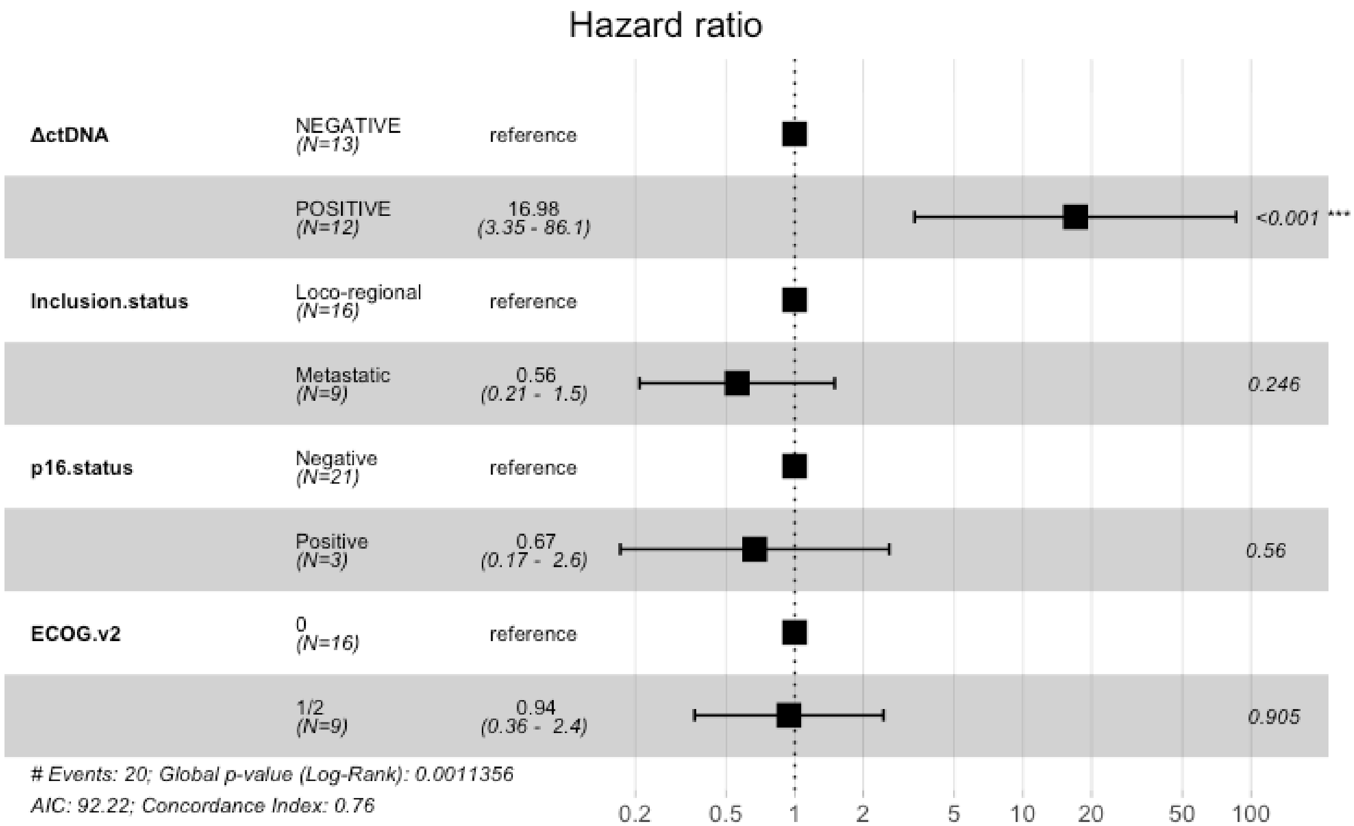
Hide

```
summary(KM_curve, times= c(0, 12, 24, 36))
```

```
Call: survfit(formula = surv_object ~ ΔctDNA, data = circ_data, conf.int = 0.95,
    conf.type = "log-log")

                ΔctDNA=NEGATIVE
 time n.risk n.event survival std.err lower 95% CI upper 95% CI
    0     13       0    1.000   0.000        1.000        1.000
   12      6       6    0.538   0.138        0.248        0.760
   24      3       2    0.359   0.139        0.117        0.613
   36      2       0    0.359   0.139        0.117        0.613

                ΔctDNA=POSITIVE
        time        n.risk       n.event      survival      std.err lower 95% CI upper 95%
CI
           0            12             0             1            0            1
1
```

Hide

```
circ_data$ΔctDNA <- factor(circ_data$ΔctDNA, levels=c("NEGATIVE","POSITIVE"))
circ_data$Inclusion.status <- factor(circ_data$Inclusion.status, levels = c("Loco-region
al", "Metastatic"))
circ_data$p16.status <- factor(circ_data$p16.status, levels = c("Negative", "Positive"))
circ_data$ECOG.v2 <- factor(circ_data$ECOG.v2, levels = c("0", "1/2"))
cox_fit <- coxph(surv_object ~ ΔctDNA + Inclusion.status + p16.status + ECOG.v2, data=ci
rc_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ ΔctDNA + Inclusion.status + p16.status +
    ECOG.v2, data = circ_data)

  n= 24, number of events= 20
   (1 observation deleted due to missingness)

                               coef exp(coef) se(coef)      z Pr(>|z|)
ΔctDNAPOSITIVE              2.83213  16.98167  0.82815  3.420 0.000627 ***
Inclusion.statusMetastatic -0.58271   0.55838  0.50257 -1.159 0.246269
p16.statusPositive         -0.40417   0.66753  0.69270 -0.583 0.559573
ECOG.v21/2                 -0.05836   0.94331  0.48669 -0.120 0.904549
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

                           exp(coef) exp(-coef) lower .95 upper .95
ΔctDNAPOSITIVE               16.9817    0.05889    3.3501    86.081
Inclusion.statusMetastatic    0.5584    1.79089    0.2085     1.495
p16.statusPositive            0.6675    1.49806    0.1717     2.595
ECOG.v21/2                    0.9433    1.06010    0.3634     2.449

Concordance= 0.758  (se = 0.048 )
Likelihood ratio test= 18.18  on 4 df,   p=0.001
Wald test            = 11.96  on 4 df,   p=0.02
Score (logrank) test = 17.74  on 4 df,   p=0.001
```

Hide

```
cox_fit_summary <- summary(cox_fit)

circ_data$ΔctDNA <- factor(circ_data$ΔctDNA, levels = c("NEGATIVE", "POSITIVE"), labels
= c("Decreased", "Increased"))
circ_data$PFS.Event <- factor(circ_data$PFS.Event, levels = c("FALSE", "TRUE"), labels =
c("No Progression", "Progression"))
contingency_table <- table(circ_data$ΔctDNA, circ_data$PFS.Event)
chi_square_test <- chisq.test(contingency_table)
```

```
 G2; H2;Warning h in stats::chisq.test(x, y, ...) :
  Chi-squared approximation may be incorrect g
```

Hide

```
print(chi_square_test)
```

```


    Pearson's Chi-squared test with Yates' continuity correction

data:  contingency_table
X-squared = 3.6158, df = 1, p-value = 0.05723
```

Hide

```
fisher_exact_test <- fisher.test(contingency_table)
print(fisher_exact_test)
```

```

	Fisher's Exact Test for Count Data

data:  contingency_table
p-value = 0.03913
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 1.013281      Inf
sample estimates:
odds ratio 
       Inf 
```

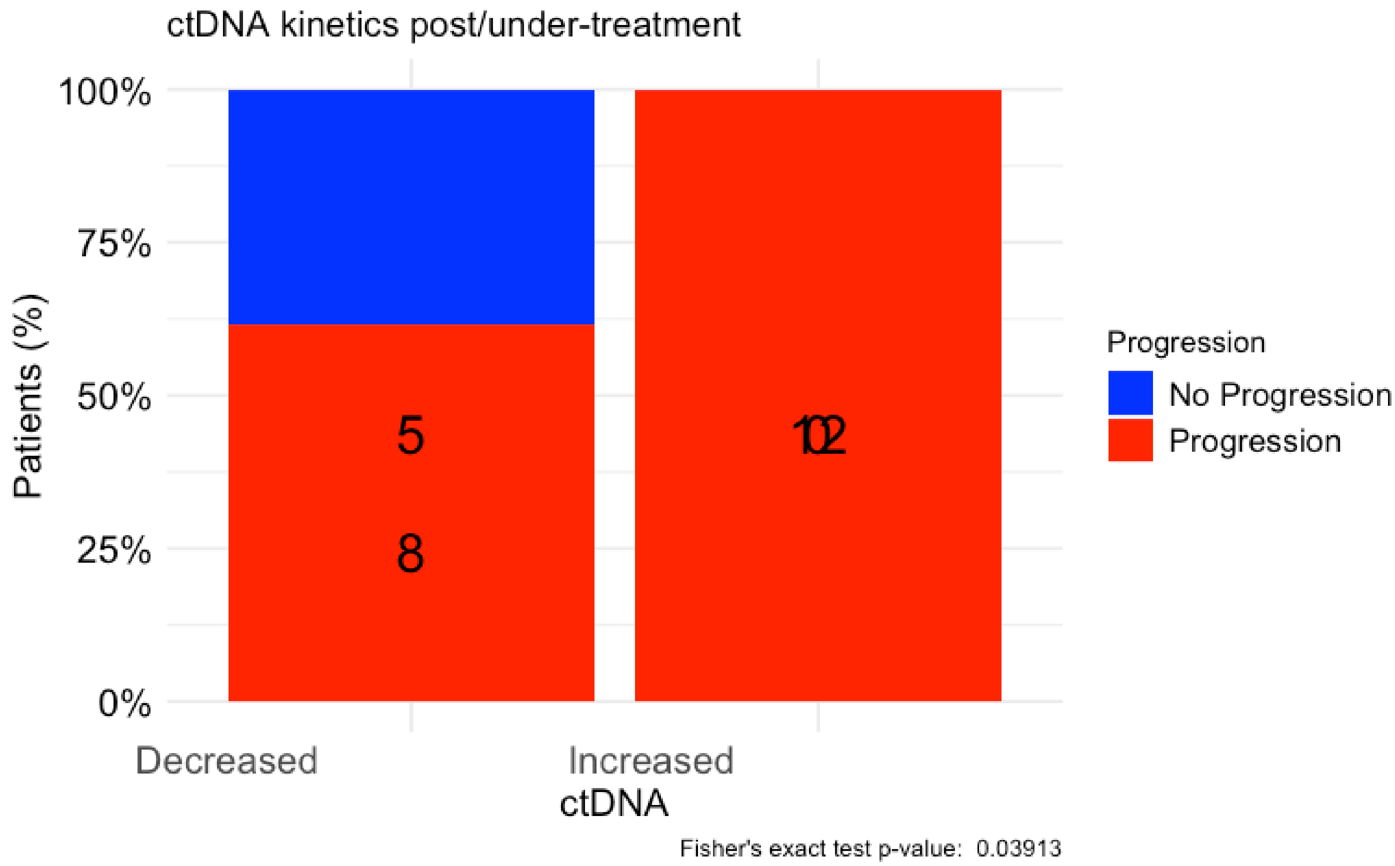
Hide

```
print(contingency_table)
```

```
           
            No Progression Progression
  Decreased              5           8
  Increased              0          12
```

Hide

```
table_df <- as.data.frame(contingency_table)
table_df$Total <- ave(table_df$Freq, table_df$Var1, FUN = sum)
table_df$Percentage <- table_df$Freq / table_df$Total
table_df$MiddlePercentage <- table_df$Percentage / 2
ggplot(table_df, aes(x = Var1, y = Percentage, fill = Var2)) +
  geom_bar(stat = "identity") +
  geom_text(aes(y = MiddlePercentage, label = Freq), position = "stack", color = "blac
k", vjust = 1.5, size = 7) +
  theme_minimal() +
  labs(title = "ctDNA kinetics post/under-treatment",
       x = "ctDNA",
       y = "Patients (%)",
       fill = "Progression",
       caption = paste("Fisher's exact test p-value: ", format.pval(fisher_exact_test$p.
value))) +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_manual(values = c("No Progression" = "blue", "Progression" = "red")) + # de
fine custom colors
  theme(axis.text.x = element_text(angle = 0, hjust = 1.5, size = 14), # increase x-axis
text size
        axis.text.y = element_text(size = 14, color = "black"), # increase y-axis text s
ize
        axis.title.x = element_text(size = 14, color = "black"), # increase x-axis label
size
        axis.title.y = element_text(size = 14, color = "black"), # increase y-axis label
size
        legend.text = element_text(size = 12, color = "black"))  # increase Progression
label size
```



#OS by ctDNA kinetics post/during-ICI

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ΔctDNA!="",]
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$OS.months, event = circ_data$OS.Event)~ΔctDNA, data = circ
_data)
```

```
Call: survfit(formula = Surv(time = circ_data$OS.months, event = circ_data$OS.Event) ~
    ΔctDNA, data = circ_data)

                 n events median 0.95LCL 0.95UCL
ΔctDNA=NEGATIVE 13      8   18.0    9.82      NA
ΔctDNA=POSITIVE 12     10   10.8    4.44      NA
```

Hide

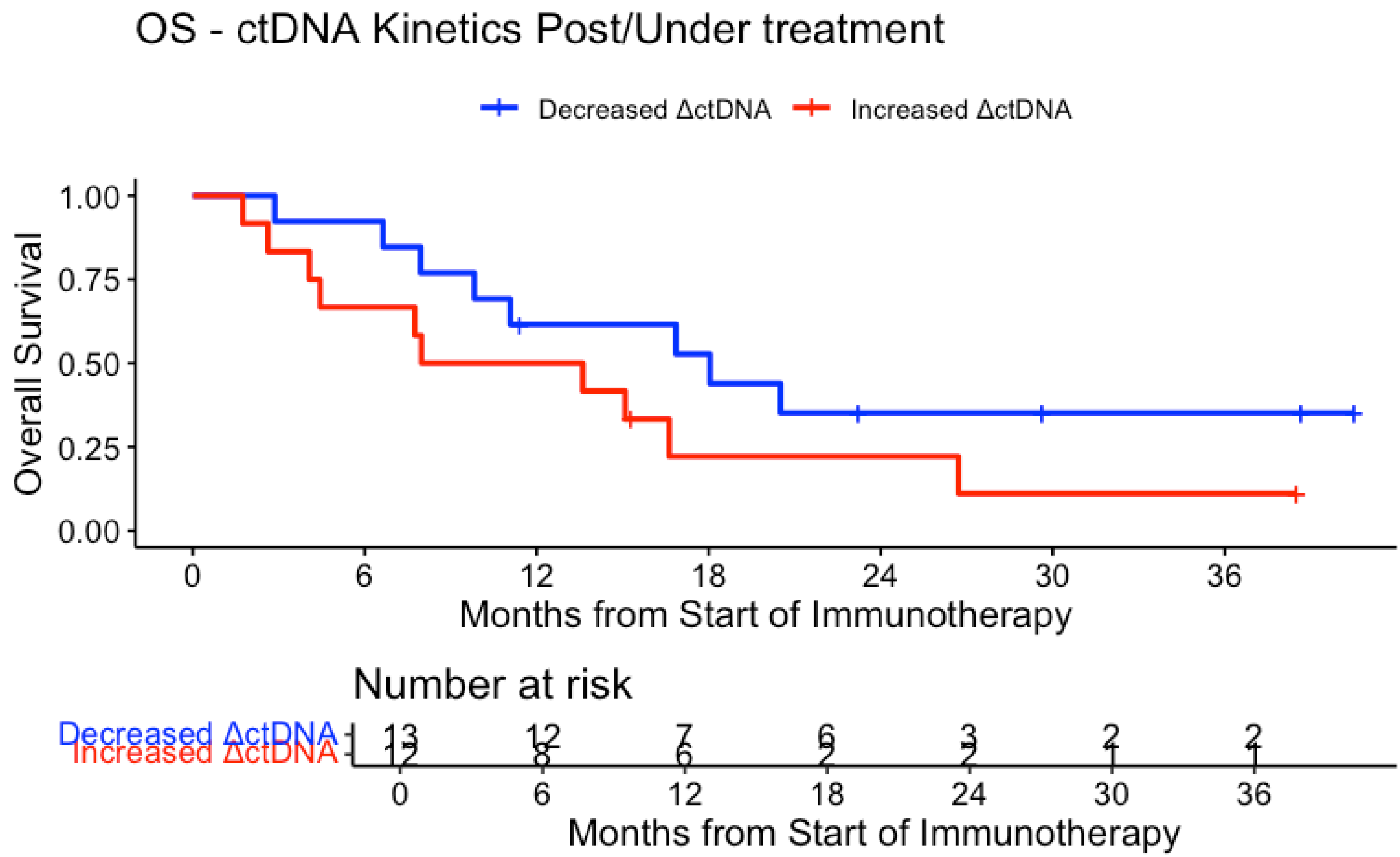
```
event_summary <- circ_data %>%
  group_by(ΔctDNA) %>%
  summarise(
    Total = n(),
    Events = sum(OS.Event),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

ΔctDNA <chr>	Total <int>	Events <int>	Fraction <dbl>	Percentage <dbl>
NEGATIVE	13	8	0.6153846	61.53846
POSITIVE	12	10	0.8333333	83.33333

2 rows

Hide

```
surv_object <-Surv(time = circ_data$OS.months, event = circ_data$OS.Event)
KM_curve <- survfit(surv_object ~ ΔctDNA, data = circ_data,conf.int=0.95,conf.type="log-
log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRU
E, break.time.by=6, palette=c("blue","red"), title="OS - ctDNA Kinetics Post/Under treat
ment", ylab= "Overall Survival", xlab="Months from Start of Immunotherapy", legend.labs=
c("Decreased ΔctDNA", "Increased ΔctDNA"), legend.title="")
```



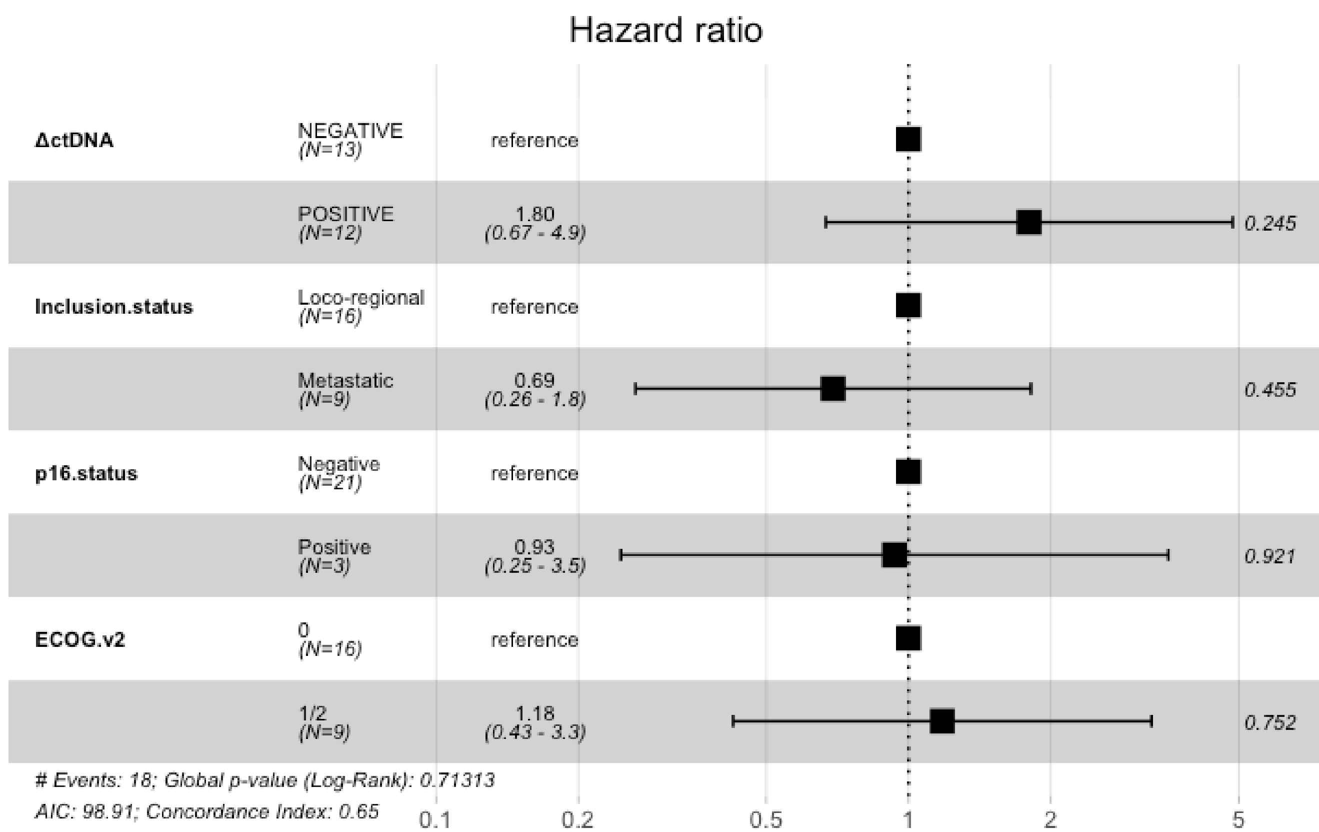
Hide

```
summary(KM_curve, times= c(0, 12, 24, 36))
```

```
Call: survfit(formula = surv_object ~ ΔctDNA, data = circ_data, conf.int = 0.95,
    conf.type = "log-log")

                ΔctDNA=NEGATIVE
 time n.risk n.event survival std.err lower 95% CI upper 95% CI
    0     13       0    1.000   0.000        1.000        1.000
   12      7       5    0.615   0.135        0.308        0.818
   24      3       3    0.352   0.139        0.112        0.607
   36      2       0    0.352   0.139        0.112        0.607

                ΔctDNA=POSITIVE
 time n.risk n.event survival std.err lower 95% CI upper 95% CI
    0     12       0    1.000   0.000      1.00000        1.000
   12      6       6    0.500   0.144      0.20848        0.736
   24      2       3    0.222   0.128      0.04111        0.492
   36      1       1    0.111   0.101      0.00701        0.378
```

Hide

```
circ_data$ΔctDNA <- factor(circ_data$ΔctDNA, levels=c("NEGATIVE","POSITIVE"))
circ_data$Inclusion.status <- factor(circ_data$Inclusion.status, levels = c("Loco-region
al", "Metastatic"))
circ_data$p16.status <- factor(circ_data$p16.status, levels = c("Negative", "Positive"))
circ_data$ECOG.v2 <- factor(circ_data$ECOG.v2, levels = c("0", "1/2"))
cox_fit <- coxph(surv_object ~ ΔctDNA + Inclusion.status + p16.status + ECOG.v2, data=ci
rc_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ ΔctDNA + Inclusion.status + p16.status +
    ECOG.v2, data = circ_data)

  n= 24, number of events= 18
   (1 observation deleted due to missingness)

                               coef exp(coef) se(coef)      z Pr(>|z|)
ΔctDNAPOSITIVE              0.58779   1.80000  0.50575  1.162    0.245
Inclusion.statusMetastatic -0.36730   0.69261  0.49166 -0.747    0.455
p16.statusPositive         -0.06789   0.93436  0.68080 -0.100    0.921
ECOG.v21/2                  0.16464   1.17897  0.52011  0.317    0.752

                           exp(coef) exp(-coef) lower .95 upper .95
ΔctDNAPOSITIVE                1.8000     0.5556    0.6680     4.850
Inclusion.statusMetastatic    0.6926     1.4438    0.2642     1.815
p16.statusPositive            0.9344     1.0702    0.2460     3.548
ECOG.v21/2                    1.1790     0.8482    0.4254     3.268

Concordance= 0.651  (se = 0.062 )
Likelihood ratio test= 2.12  on 4 df,   p=0.7
Wald test            = 2.1  on 4 df,   p=0.7
Score (logrank) test = 2.15  on 4 df,   p=0.7
```

Hide

```
cox_fit_summary <- summary(cox_fit)

circ_data$ΔctDNA <- factor(circ_data$ΔctDNA, levels = c("NEGATIVE", "POSITIVE"), labels
= c("Decreased", "Increased"))
circ_data$OS.Event <- factor(circ_data$OS.Event, levels = c("FALSE", "TRUE"), labels = c
("Alive", "Deceased"))
contingency_table <- table(circ_data$ΔctDNA, circ_data$OS.Event)
chi_square_test <- chisq.test(contingency_table)
```

```
 G2; H2;Warning h in stats::chisq.test(x, y, ...) :
  Chi-squared approximation may be incorrect g
```

Hide

```
print(chi_square_test)
```

```


    Pearson's Chi-squared test with Yates' continuity correction

data:  contingency_table
X-squared = 0.58792, df = 1, p-value = 0.4432
```

Hide

```
fisher_exact_test <- fisher.test(contingency_table)
print(fisher_exact_test)
```

```

	Fisher's Exact Test for Count Data

data:  contingency_table
p-value = 0.3783
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
  0.363132 39.390335
sample estimates:
odds ratio 
  2.985191 
```

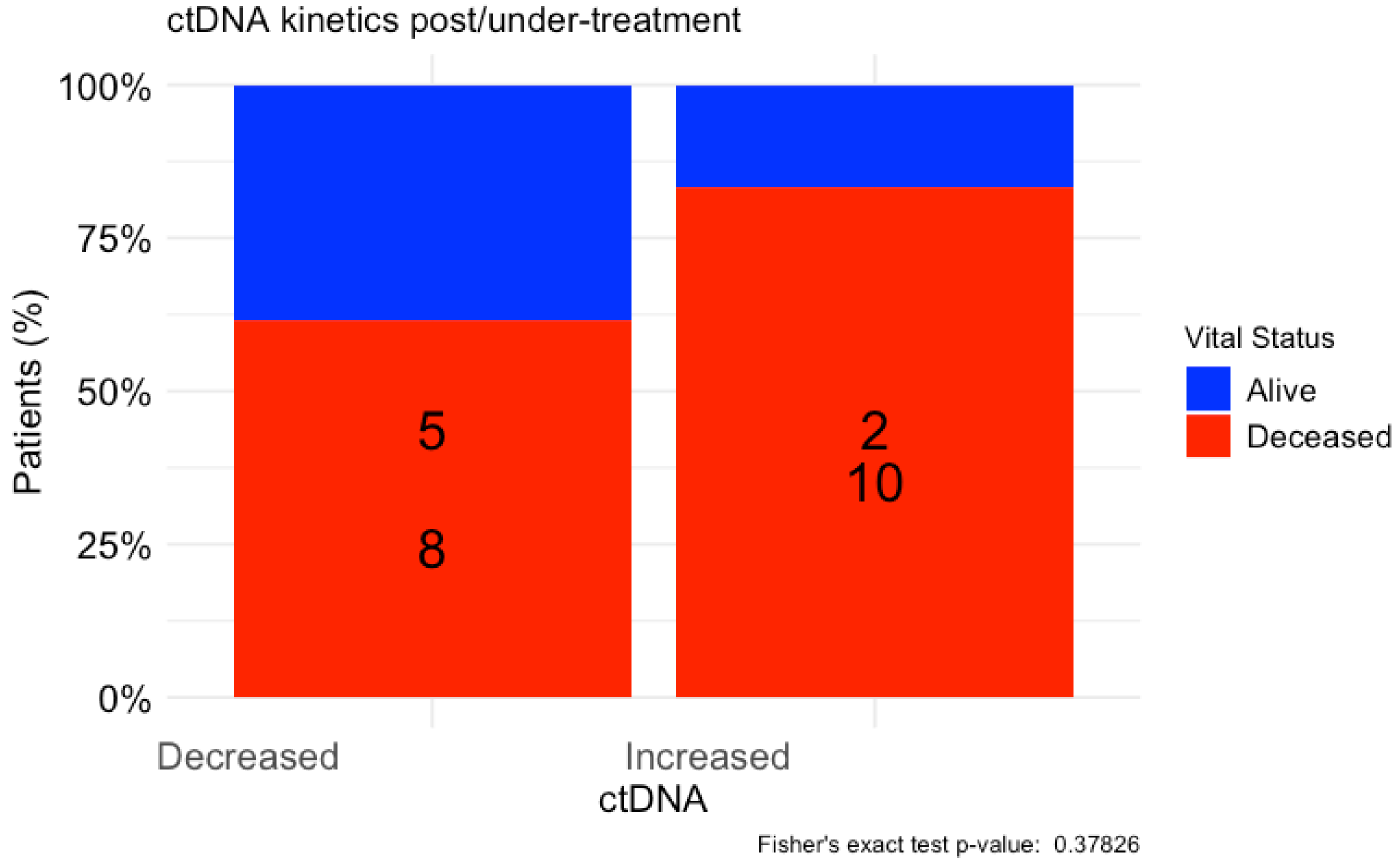
Hide

```
print(contingency_table)
```

```
           
            Alive Deceased
  Decreased     5        8
  Increased     2       10
```

Hide

```
table_df <- as.data.frame(contingency_table)
table_df$Total <- ave(table_df$Freq, table_df$Var1, FUN = sum)
table_df$Percentage <- table_df$Freq / table_df$Total
table_df$MiddlePercentage <- table_df$Percentage / 2
ggplot(table_df, aes(x = Var1, y = Percentage, fill = Var2)) +
  geom_bar(stat = "identity") +
  geom_text(aes(y = MiddlePercentage, label = Freq), position = "stack", color = "blac
k", vjust = 1.5, size = 7) +
  theme_minimal() +
  labs(title = "ctDNA kinetics post/under-treatment",
       x = "ctDNA",
       y = "Patients (%)",
       fill = "Vital Status",
       caption = paste("Fisher's exact test p-value: ", format.pval(fisher_exact_test$p.
value))) +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_manual(values = c("Alive" = "blue", "Deceased" = "red")) + # define custom
colors
  theme(axis.text.x = element_text(angle = 0, hjust = 1.5, size = 14), # increase x-axis
text size
        axis.text.y = element_text(size = 14, color = "black"), # increase y-axis text s
ize
        axis.title.x = element_text(size = 14, color = "black"), # increase x-axis label
size
        axis.title.y = element_text(size = 14, color = "black"), # increase y-axis label
size
        legend.text = element_text(size = 12, color = "black"))  # increase Progression
label size
```



#Association of ctDNA kinetics post/during-ICI with BOR

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ΔctDNA!="",]
circ_datadf <- as.data.frame(circ_data)

circ_data$ΔctDNA <- factor(circ_data$ΔctDNA, levels = c("NEGATIVE", "POSITIVE"), labels
= c("Decreased", "Increased"))
circ_data$RESIST <- factor(circ_data$RESIST, levels = c("CR", "PR", "SD", "PD"))
contingency_table <- table(circ_data$ΔctDNA, circ_data$RESIST)
chi_square_test <- chisq.test(contingency_table)
```

```
G2; H2;Warning h in stats::chisq.test(x, y, ...) :
  Chi-squared approximation may be incorrect g
```

Hide

```
print(chi_square_test)
```

```
	Pearson's Chi-squared test

data:  contingency_table
X-squared = 15.385, df = 3, p-value = 0.001516
```

Hide

```
fisher_exact_test <- fisher.test(contingency_table)
print(fisher_exact_test)
```

```
	Fisher's Exact Test for Count Data

data:  contingency_table
p-value = 0.0003154
alternative hypothesis: two.sided
```

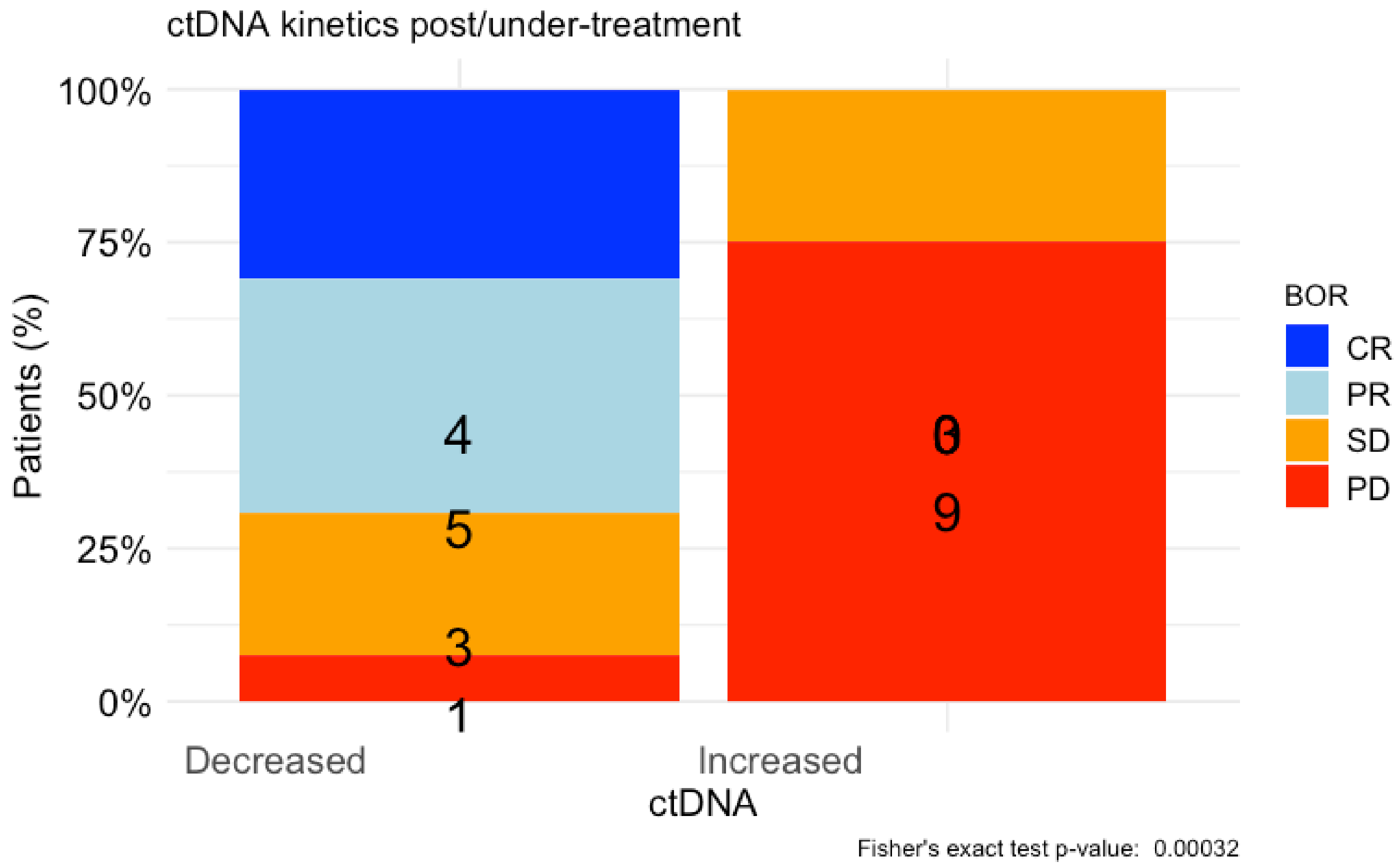
Hide

```
print(contingency_table)
```

```
            CR PR SD PD
  Decreased  4  5  3  1
  Increased  0  0  3  9
```

Hide

```
table_df <- as.data.frame(contingency_table)
table_df$Total <- ave(table_df$Freq, table_df$Var1, FUN = sum)
table_df$Percentage <- table_df$Freq / table_df$Total
table_df$MiddlePercentage <- table_df$Percentage / 2
ggplot(table_df, aes(x = Var1, y = Percentage, fill = Var2)) +
  geom_bar(stat = "identity") +
  geom_text(aes(y = MiddlePercentage, label = Freq), position = "stack", color = "blac
k", vjust = 1.5, size = 7) +
  theme_minimal() +
  labs(title = "ctDNA kinetics post/under-treatment",
       x = "ctDNA",
       y = "Patients (%)",
       fill = "BOR",
       caption = paste("Fisher's exact test p-value: ", format.pval(fisher_exact_test$p.
value))) +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_manual(values = c("CR" = "blue", "PR" = "lightblue", "SD" = "orange", "PD"
= "red")) + # define custom colors
  theme(axis.text.x = element_text(angle = 0, hjust = 1.5, size = 14), # increase x-axis
text size
        axis.text.y = element_text(size = 14, color = "black"), # increase y-axis text s
ize
        axis.title.x = element_text(size = 14, color = "black"), # increase x-axis label
size
        axis.title.y = element_text(size = 14, color = "black"), # increase y-axis label
size
        legend.text = element_text(size = 12, color = "black"))  # increase Progression
label size
```



#PFS by ctDNA clearance post/during-ICI

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data <- circ_data[circ_data$ctDNA.Base=="POSITIVE",]
circ_datadf <- as.data.frame(circ_data)

circ_data$ctDNA.Dynamics <- NA
circ_data <- circ_data %>%
  mutate(ctDNA.Dynamics = case_when(
    ctDNA.Base == "POSITIVE" & ctDNA.postTx == "NEGATIVE" ~ 1,
    ctDNA.Base == "POSITIVE" & ctDNA.postTx == "POSITIVE" ~ 2
  ))

survfit(Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)~ctDNA.Dynamics, d
ata = circ_data)
```

```
Call: survfit(formula = Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event) ~
    ctDNA.Dynamics, data = circ_data)

                  n events median 0.95LCL 0.95UCL
ctDNA.Dynamics=1  6      2     NA   19.68      NA
ctDNA.Dynamics=2 19     18   3.68    2.37    7.95
```

Hide

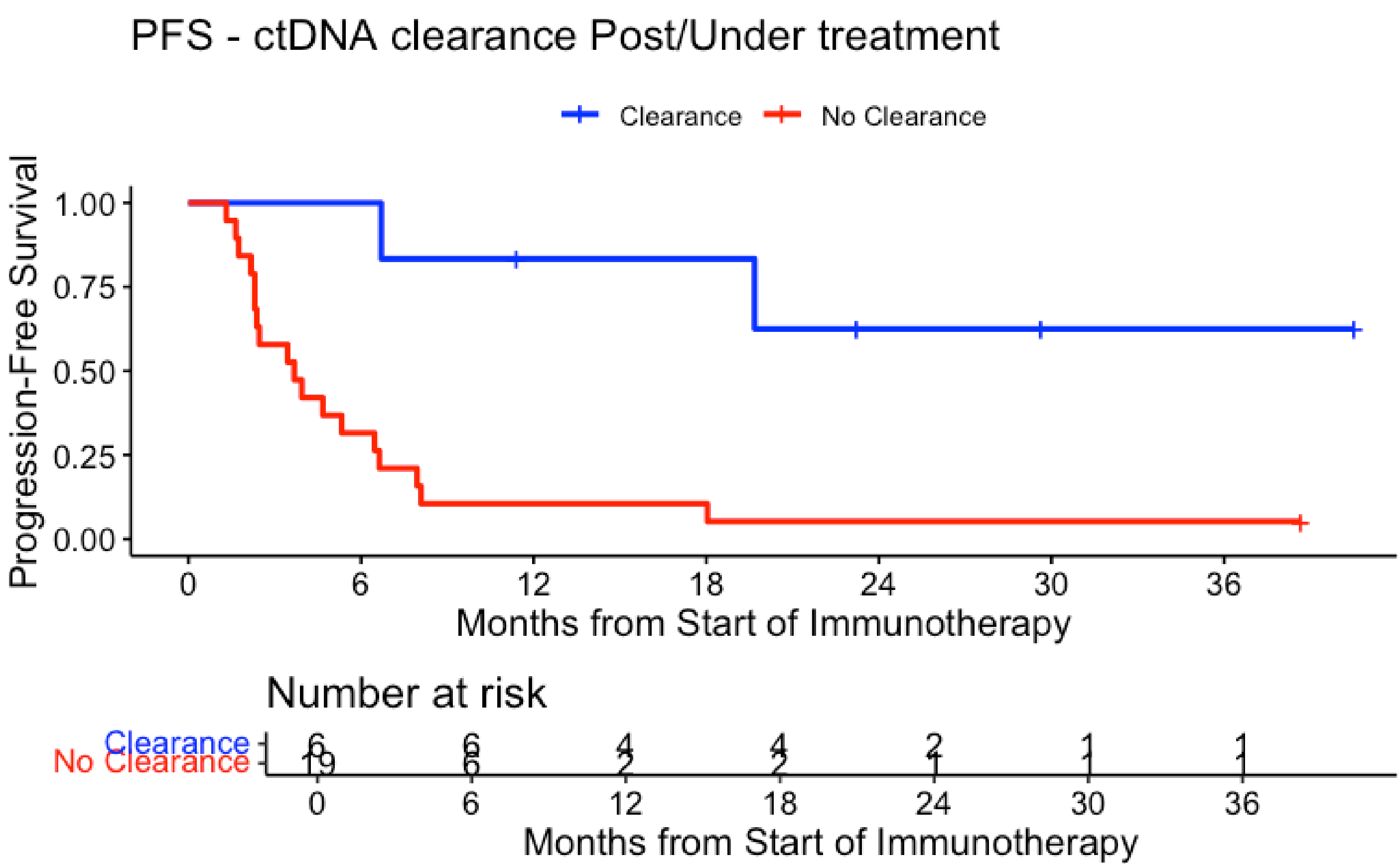
```
event_summary <- circ_data %>%
  group_by(ctDNA.Dynamics) %>%
  summarise(
    Total = n(),
    Events = sum(PFS.Event),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

ctDNA.Dynamics <dbl>	Total <int>	Events <int>	Fraction <dbl>	Percentage <dbl>
1	6	2	0.3333333	33.33333
2	19	18	0.9473684	94.73684

2 rows

Hide

```
surv_object <-Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.Dynamics, data = circ_data,conf.int=0.95,conf.ty
pe="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRU
E, break.time.by=6, palette=c("blue","red"), title="PFS - ctDNA clearance Post/Under tre
atment", ylab= "Progression-Free Survival", xlab="Months from Start of Immunotherapy", l
egend.labs=c("Clearance", "No Clearance"), legend.title="")
```



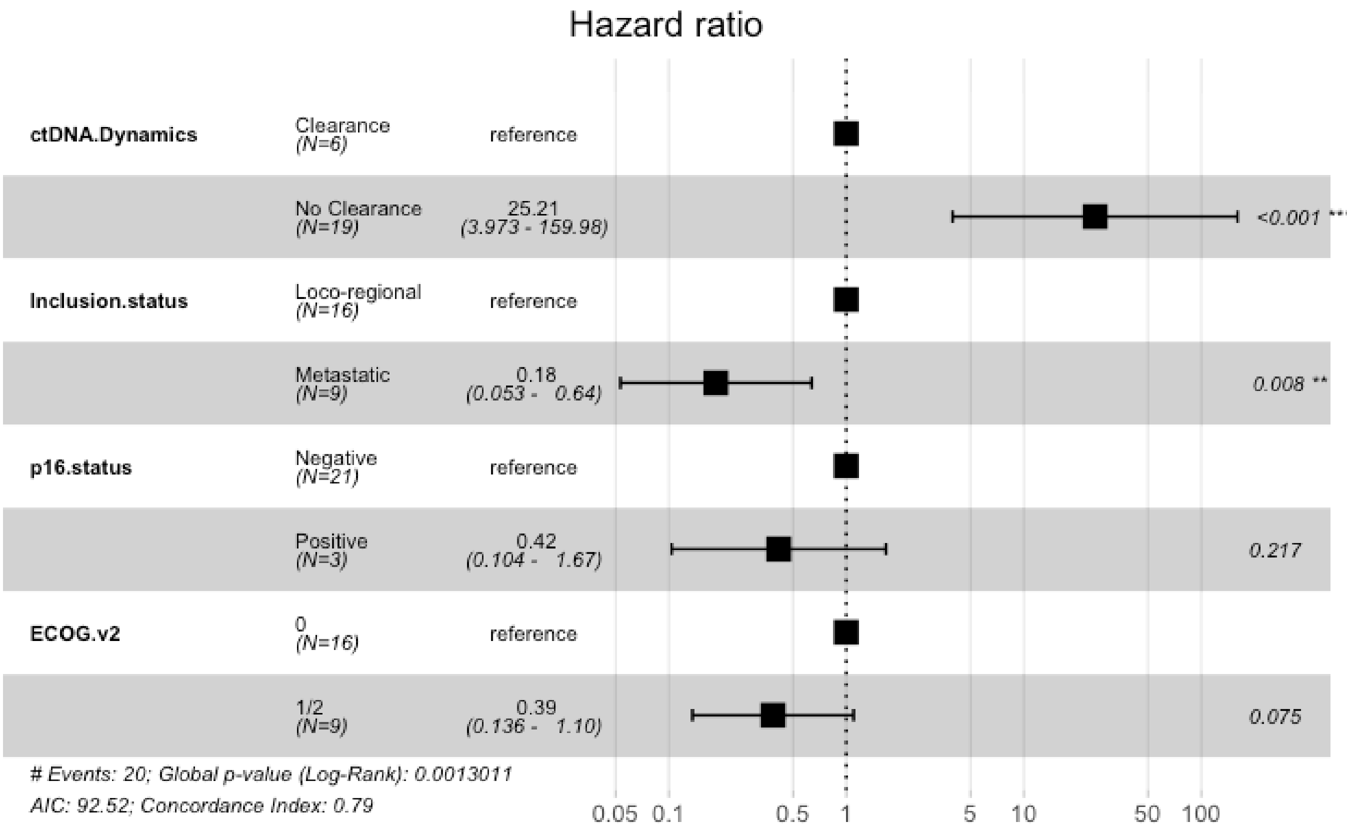
Hide

```
summary(KM_curve, times= c(0, 12, 24, 36))
```

```
Call: survfit(formula = surv_object ~ ctDNA.Dynamics, data = circ_data, 
    conf.int = 0.95, conf.type = "log-log")

                ctDNA.Dynamics=1 
 time n.risk n.event survival std.err lower 95% CI upper 95% CI
    0      6       0    1.000   0.000        1.000        1.000
   12      4       1    0.833   0.152        0.273        0.975
   24      2       1    0.625   0.213        0.142        0.893
   36      1       0    0.625   0.213        0.142        0.893

                ctDNA.Dynamics=2 
 time n.risk n.event survival std.err lower 95% CI upper 95% CI
    0     19       0   1.0000  0.0000      1.00000        1.000
   12      2      17   0.1053  0.0704      0.01777        0.284
   24      1       1   0.0526  0.0512      0.00359        0.214
   36      1       0   0.0526  0.0512      0.00359        0.214
```

Hide

```
circ_data$ctDNA.Dynamics <- factor(circ_data$ctDNA.Dynamics, levels=c("1","2"), labels = 
c("Clearance", "No Clearance"))
circ_data$Inclusion.status <- factor(circ_data$Inclusion.status, levels = c("Loco-region
al", "Metastatic"))
circ_data$p16.status <- factor(circ_data$p16.status, levels = c("Negative", "Positive"))
circ_data$ECOG.v2 <- factor(circ_data$ECOG.v2, levels = c("0", "1/2"))
cox_fit <- coxph(surv_object ~ ctDNA.Dynamics + Inclusion.status + p16.status + ECOG.v2, 
data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ ctDNA.Dynamics + Inclusion.status + 
    p16.status + ECOG.v2, data = circ_data)

  n= 24, number of events= 20 
   (1 observation deleted due to missingness)

                              coef exp(coef) se(coef)      z Pr(>|z|)    
ctDNA.DynamicsNo Clearance  3.2273   25.2118   0.9427  3.423 0.000618 ***
Inclusion.statusMetastatic -1.6899    0.1845   0.6328 -2.671 0.007568 ** 
p16.statusPositive         -0.8753    0.4167   0.7088 -1.235 0.216873    
ECOG.v21/2                 -0.9498    0.3868   0.5328 -1.783 0.074662 .  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

                           exp(coef) exp(-coef) lower .95 upper .95
ctDNA.DynamicsNo Clearance   25.2118    0.03966   3.97331  159.9761
Inclusion.statusMetastatic    0.1845    5.41913   0.05339    0.6378
p16.statusPositive            0.4167    2.39956   0.10388    1.6719
ECOG.v21/2                    0.3868    2.58524   0.13613    1.0992

Concordance= 0.787  (se = 0.036 )
Likelihood ratio test= 17.88  on 4 df,   p=0.001
Wald test            = 13.08  on 4 df,   p=0.01
Score (logrank) test = 15.23  on 4 df,   p=0.004
```

Hide

```
cox_fit_summary <- summary(cox_fit)

circ_data$PFS.Event <- factor(circ_data$PFS.Event, levels = c("FALSE", "TRUE"), labels = c("No Progression", "Progression"))
contingency_table <- table(circ_data$ctDNA.Dynamics, circ_data$PFS.Event)
chi_square_test <- chisq.test(contingency_table)
```

```
 G2; H2;Warning h in stats::chisq.test(x, y, ...) :
  Chi-squared approximation may be incorrect g
```

Hide

```
print(chi_square_test)
```

```


    Pearson's Chi-squared test with Yates' continuity correction

data:  contingency_table
X-squared = 7.2505, df = 1, p-value = 0.007088
```

Hide

```
fisher_exact_test <- fisher.test(contingency_table)
print(fisher_exact_test)
```

```


    Fisher's Exact Test for Count Data

data:  contingency_table
p-value = 0.005477
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
    1.796359 1833.444857
sample estimates:
odds ratio
  27.59073
```

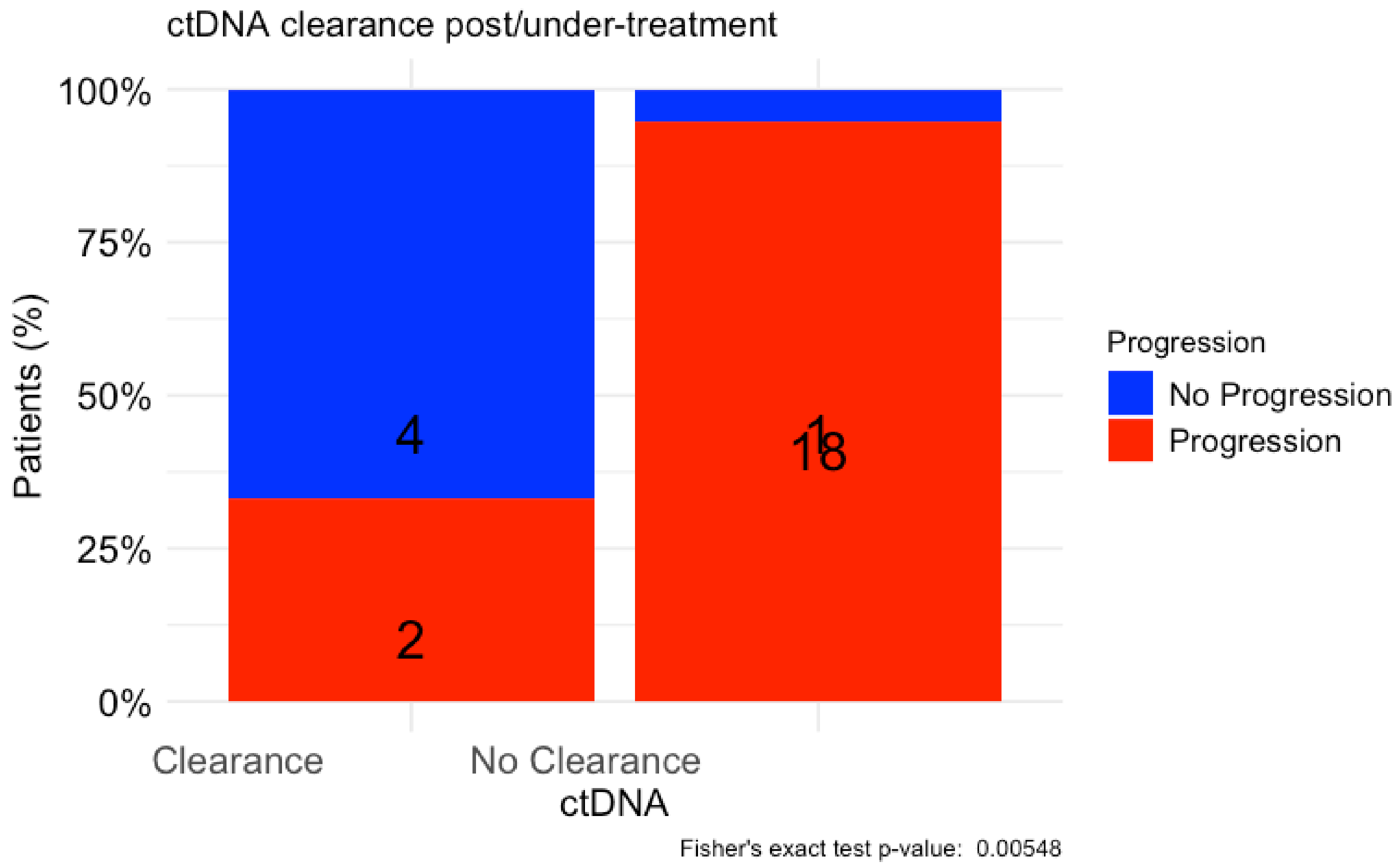
Hide

```
print(contingency_table)
```

```


               No Progression Progression
  Clearance                 4           2
  No Clearance              1          18
```

Hide

```
table_df <- as.data.frame(contingency_table)
table_df$Total <- ave(table_df$Freq, table_df$Var1, FUN = sum)
table_df$Percentage <- table_df$Freq / table_df$Total
table_df$MiddlePercentage <- table_df$Percentage / 2
ggplot(table_df, aes(x = Var1, y = Percentage, fill = Var2)) +
  geom_bar(stat = "identity") +
  geom_text(aes(y = MiddlePercentage, label = Freq), position = "stack", color = "blac
k", vjust = 1.5, size = 7) +
  theme_minimal() +
  labs(title = "ctDNA clearance post/under-treatment",
       x = "ctDNA",
       y = "Patients (%)",
       fill = "Progression",
       caption = paste("Fisher's exact test p-value: ", format.pval(fisher_exact_test$p.
value))) +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_manual(values = c("No Progression" = "blue", "Progression" = "red")) + # de
fine custom colors
  theme(axis.text.x = element_text(angle = 0, hjust = 1.5, size = 14), # increase x-axis
text size
        axis.text.y = element_text(size = 14, color = "black"), # increase y-axis text s
ize
        axis.title.x = element_text(size = 14, color = "black"), # increase x-axis label
size
        axis.title.y = element_text(size = 14, color = "black"), # increase y-axis label
size
        legend.text = element_text(size = 12, color = "black"))  # increase Progression
label size
```



#OS by ctDNA clearance post/during-ICI

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data <- circ_data[circ_data$ctDNA.Base=="POSITIVE",]
circ_datadf <- as.data.frame(circ_data)

circ_data$ctDNA.Dynamics <- NA
circ_data <- circ_data %>%
  mutate(ctDNA.Dynamics = case_when(
    ctDNA.Base == "POSITIVE" & ctDNA.postTx == "NEGATIVE" ~ 1,
    ctDNA.Base == "POSITIVE" & ctDNA.postTx == "POSITIVE" ~ 2
  ))

survfit(Surv(time = circ_data$OS.months, event = circ_data$OS.Event)~ctDNA.Dynamics, dat
a = circ_data)
```

```
Call: survfit(formula = Surv(time = circ_data$OS.months, event = circ_data$OS.Event) ~
    ctDNA.Dynamics, data = circ_data)

                  n events median 0.95LCL 0.95UCL
ctDNA.Dynamics=1  6      2     NA   20.50      NA
ctDNA.Dynamics=2 19     16   9.82    7.75    26.7
```

Hide

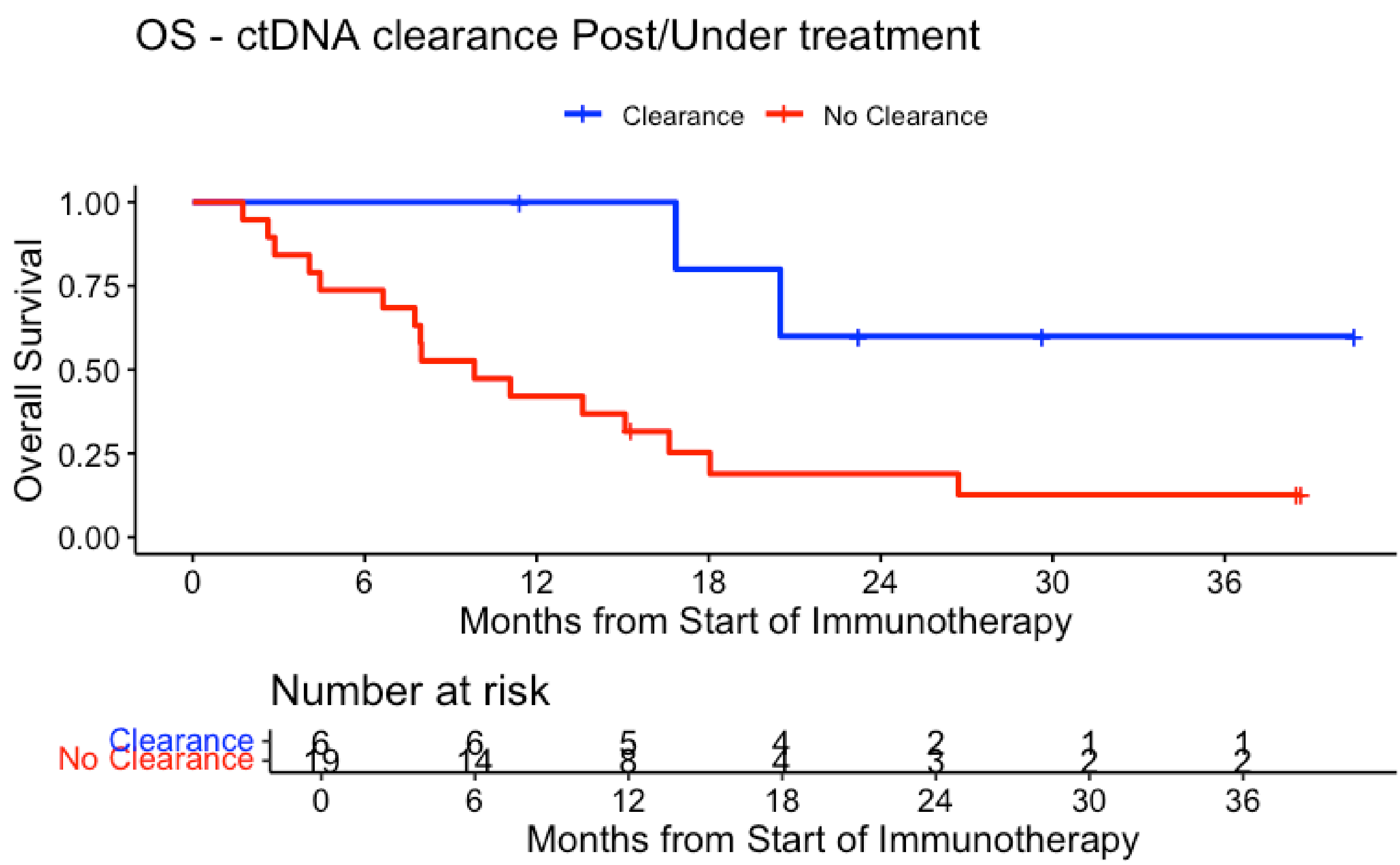
```
event_summary <- circ_data %>%
  group_by(ctDNA.Dynamics) %>%
  summarise(
    Total = n(),
    Events = sum(OS.Event),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

ctDNA.Dynamics <dbl>	Total <int>	Events <int>	Fraction <dbl>	Percentage <dbl>
1	6	2	0.3333333	33.33333
2	19	16	0.8421053	84.21053

2 rows

Hide

```
surv_object <-Surv(time = circ_data$OS.months, event = circ_data$OS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.Dynamics, data = circ_data,conf.int=0.95,conf.ty
pe="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRU
E, break.time.by=6, palette=c("blue","red"), title="OS - ctDNA clearance Post/Under trea
tment", ylab= "Overall Survival", xlab="Months from Start of Immunotherapy", legend.labs
=c("Clearance", "No Clearance"), legend.title="")
```



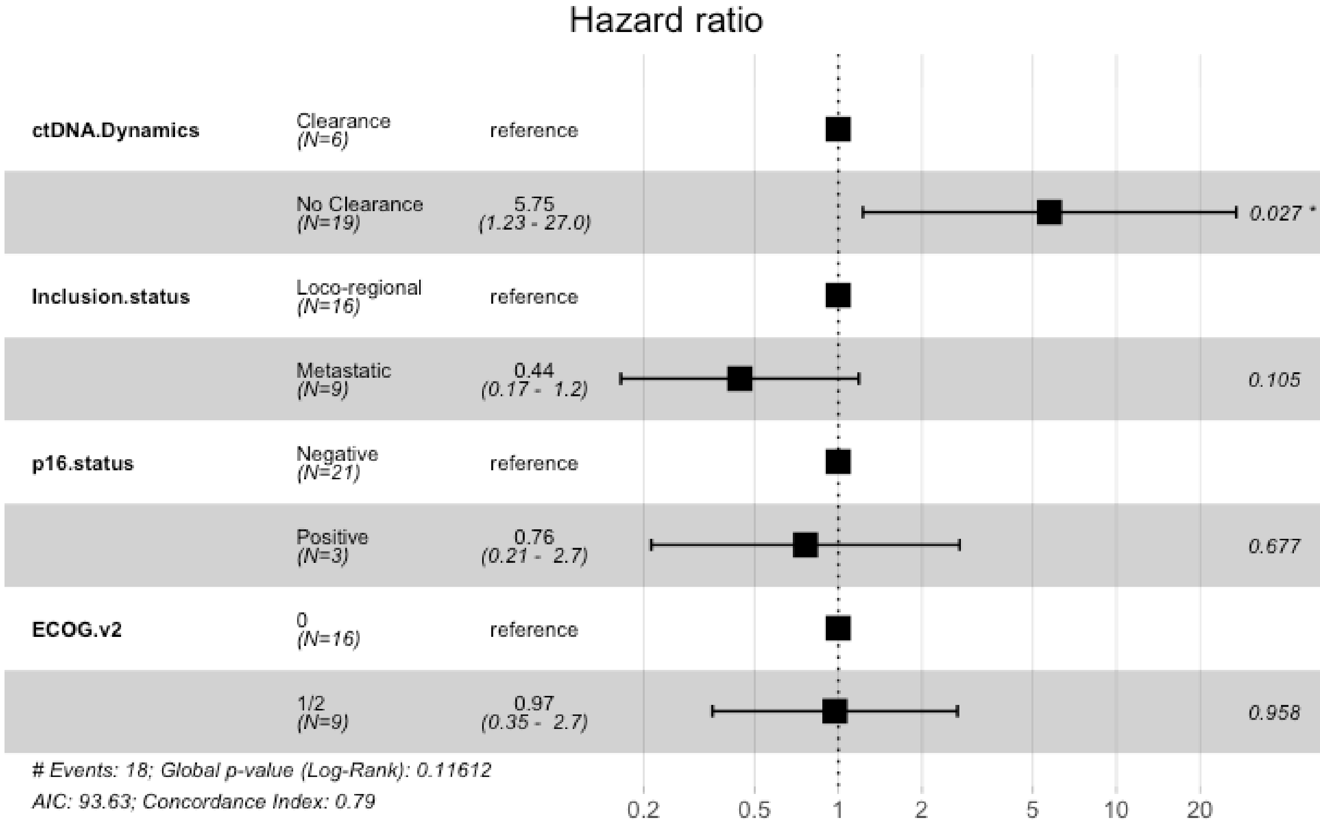
Hide

```
summary(KM_curve, times= c(0, 12, 24, 36))
```

```
Call: survfit(formula = surv_object ~ ctDNA.Dynamics, data = circ_data,
    conf.int = 0.95, conf.type = "log-log")

                ctDNA.Dynamics=1
 time n.risk n.event survival std.err lower 95% CI upper 95% CI
    0      6       0      1.0   0.000        1.000        1.000
   12      5       0      1.0   0.000           NA           NA
   24      2       2      0.6   0.219        0.126        0.882
   36      1       0      0.6   0.219        0.126        0.882

                ctDNA.Dynamics=2
 time n.risk n.event survival std.err lower 95% CI upper 95% CI
    0     19       0    1.000  0.0000       1.0000        1.000
   12      8      11    0.421  0.1133       0.2037        0.625
   24      3       4    0.189  0.0942       0.0503        0.396
   36      2       1    0.126  0.0813       0.0222        0.325
```

Hide

```
circ_data$ctDNA.Dynamics <- factor(circ_data$ctDNA.Dynamics, levels=c("1","2"), labels =
c("Clearance", "No Clearance"))
circ_data$Inclusion.status <- factor(circ_data$Inclusion.status, levels = c("Loco-region
al", "Metastatic"))
circ_data$p16.status <- factor(circ_data$p16.status, levels = c("Negative", "Positive"))
circ_data$ECOG.v2 <- factor(circ_data$ECOG.v2, levels = c("0", "1/2"))
cox_fit <- coxph(surv_object ~ ctDNA.Dynamics + Inclusion.status + p16.status + ECOG.v2,
data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ ctDNA.Dynamics + Inclusion.status +
    p16.status + ECOG.v2, data = circ_data)

  n= 24, number of events= 18
   (1 observation deleted due to missingness)

                              coef exp(coef) se(coef)      z Pr(>|z|)
ctDNA.DynamicsNo Clearance  1.7490    5.7485   0.7885  2.218   0.0265 *
Inclusion.statusMetastatic -0.8151    0.4426   0.5028 -1.621   0.1050
p16.statusPositive         -0.2716    0.7622   0.6514 -0.417   0.6767
ECOG.v21/2                 -0.0275    0.9729   0.5174 -0.053   0.9576
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

                           exp(coef) exp(-coef) lower .95 upper .95
ctDNA.DynamicsNo Clearance    5.7486      0.174    1.2258    26.958
Inclusion.statusMetastatic    0.4426      2.259    0.1652     1.186
p16.statusPositive            0.7622      1.312    0.2126     2.732
ECOG.v21/2                    0.9729      1.028    0.3529     2.682

Concordance= 0.788  (se = 0.061 )
Likelihood ratio test= 7.4  on 4 df,   p=0.1
Wald test            = 6.28  on 4 df,   p=0.2
Score (logrank) test = 6.92  on 4 df,   p=0.1
```

Hide

```
cox_fit_summary <- summary(cox_fit)

circ_data$OS.Event <- factor(circ_data$OS.Event, levels = c("FALSE", "TRUE"), labels = c
("Alive", "Deceased"))
contingency_table <- table(circ_data$ctDNA.Dynamics, circ_data$OS.Event)
chi_square_test <- chisq.test(contingency_table)
```

```
G2; H2;Warning h in stats::chisq.test(x, y, ...) :
  Chi-squared approximation may be incorrect g
```

Hide

```
print(chi_square_test)
```

```
    Pearson's Chi-squared test with Yates' continuity correction

data:  contingency_table
X-squared = 3.6032, df = 1, p-value = 0.05767
```

Hide

```
fisher_exact_test <- fisher.test(contingency_table)
print(fisher_exact_test)
```

```

	Fisher's Exact Test for Count Data

data:  contingency_table
p-value = 0.03241
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
   0.8988842 150.9697807
sample estimates:
odds ratio
   9.34674
```

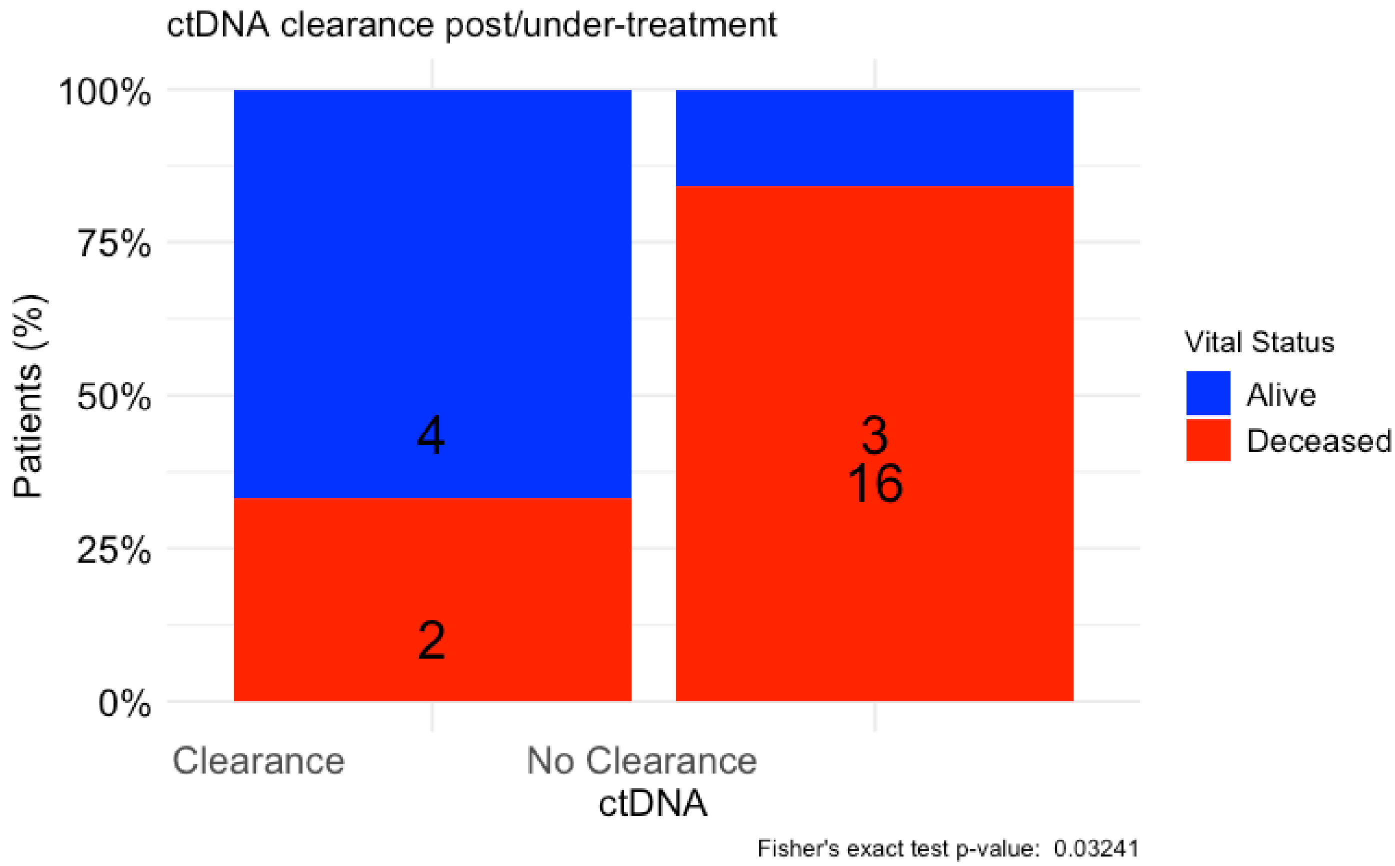
Hide

```
print(contingency_table)
```

```

               Alive Deceased
  Clearance        4        2
  No Clearance     3       16
```

Hide

```
table_df <- as.data.frame(contingency_table)
table_df$Total <- ave(table_df$Freq, table_df$Var1, FUN = sum)
table_df$Percentage <- table_df$Freq / table_df$Total
table_df$MiddlePercentage <- table_df$Percentage / 2
ggplot(table_df, aes(x = Var1, y = Percentage, fill = Var2)) +
  geom_bar(stat = "identity") +
  geom_text(aes(y = MiddlePercentage, label = Freq), position = "stack", color = "blac
k", vjust = 1.5, size = 7) +
  theme_minimal() +
  labs(title = "ctDNA clearance post/under-treatment",
       x = "ctDNA",
       y = "Patients (%)",
       fill = "Vital Status",
       caption = paste("Fisher's exact test p-value: ", format.pval(fisher_exact_test$p.
value))) +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_manual(values = c("Alive" = "blue", "Deceased" = "red")) + # define custom
colors
  theme(axis.text.x = element_text(angle = 0, hjust = 1.5, size = 14), # increase x-axis
text size
        axis.text.y = element_text(size = 14, color = "black"), # increase y-axis text s
ize
        axis.title.x = element_text(size = 14, color = "black"), # increase x-axis label
size
        axis.title.y = element_text(size = 14, color = "black"), # increase y-axis label
size
        legend.text = element_text(size = 12, color = "black"))  # increase Progression
label size
```



#Association of ctDNA clearance post/during-ICI with BOR

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data <- circ_data[circ_data$ctDNA.Base=="POSITIVE",]
circ_datadf <- as.data.frame(circ_data)

circ_data$ctDNA.Dynamics <- NA
circ_data <- circ_data %>%
  mutate(ctDNA.Dynamics = case_when(
    ctDNA.Base == "POSITIVE" & ctDNA.postTx == "NEGATIVE" ~ 1,
    ctDNA.Base == "POSITIVE" & ctDNA.postTx == "POSITIVE" ~ 2
  ))

circ_data$ctDNA.Dynamics <- factor(circ_data$ctDNA.Dynamics, levels=c("1","2"), labels =
c("Clearance", "No Clearance"))
circ_data$RESIST <- factor(circ_data$RESIST, levels = c("CR", "PR", "SD", "PD"))
contingency_table <- table(circ_data$ctDNA.Dynamics, circ_data$RESIST)
chi_square_test <- chisq.test(contingency_table)
```

```
G2; H2;Warning h in stats::chisq.test(x, y, ...) :
  Chi-squared approximation may be incorrect g
```

Hide

```
print(chi_square_test)
```

```

	Pearson's Chi-squared test

data:  contingency_table
X-squared = 14.309, df = 3, p-value = 0.002513
```

Hide

```
fisher_exact_test <- fisher.test(contingency_table)
print(fisher_exact_test)
```

```

	Fisher's Exact Test for Count Data

data:  contingency_table
p-value = 0.001129
alternative hypothesis: two.sided
```

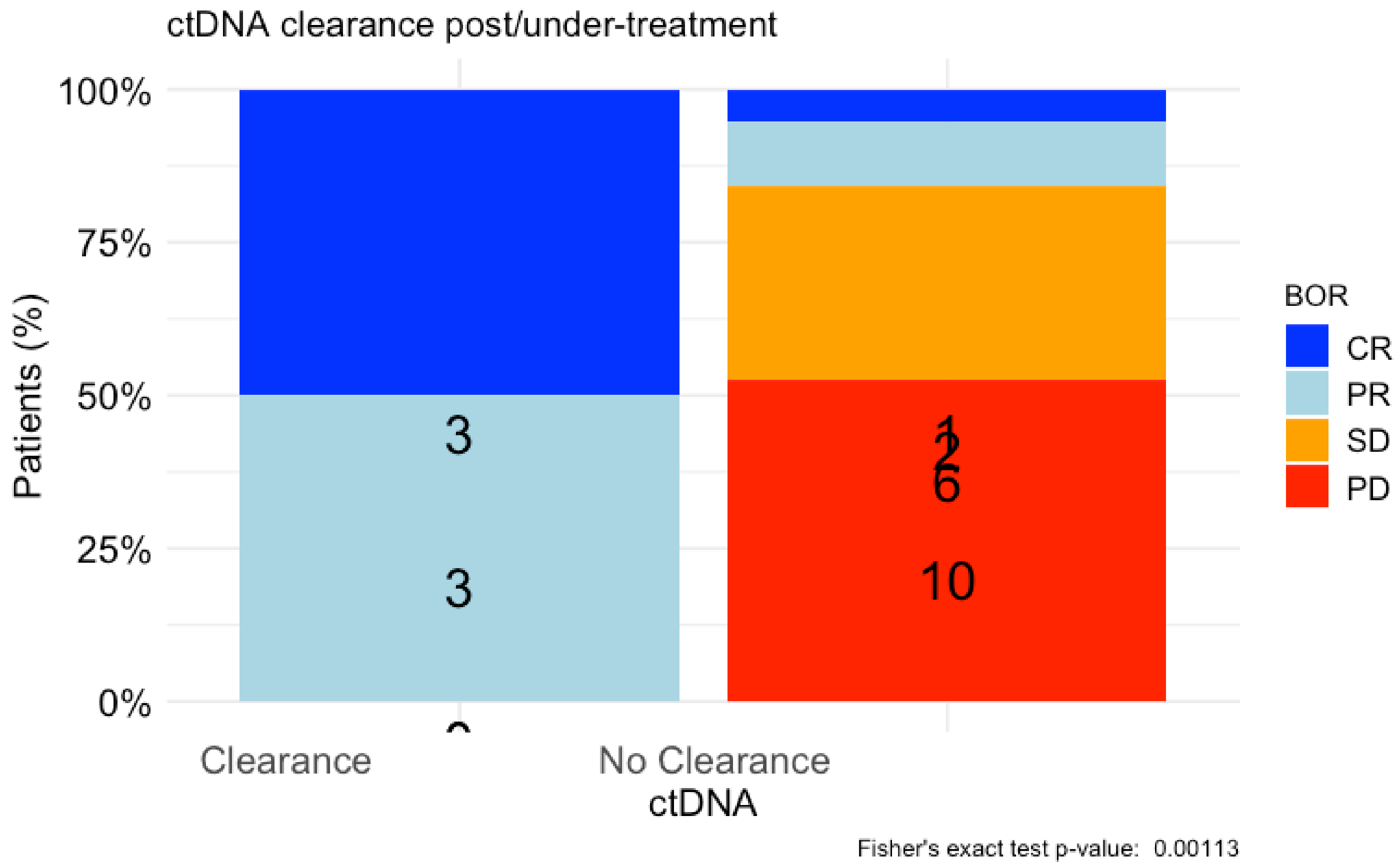
Hide

```
print(contingency_table)
```

```

               CR PR SD PD
  Clearance     3  3  0  0
  No Clearance  1  2  6 10
```

Hide

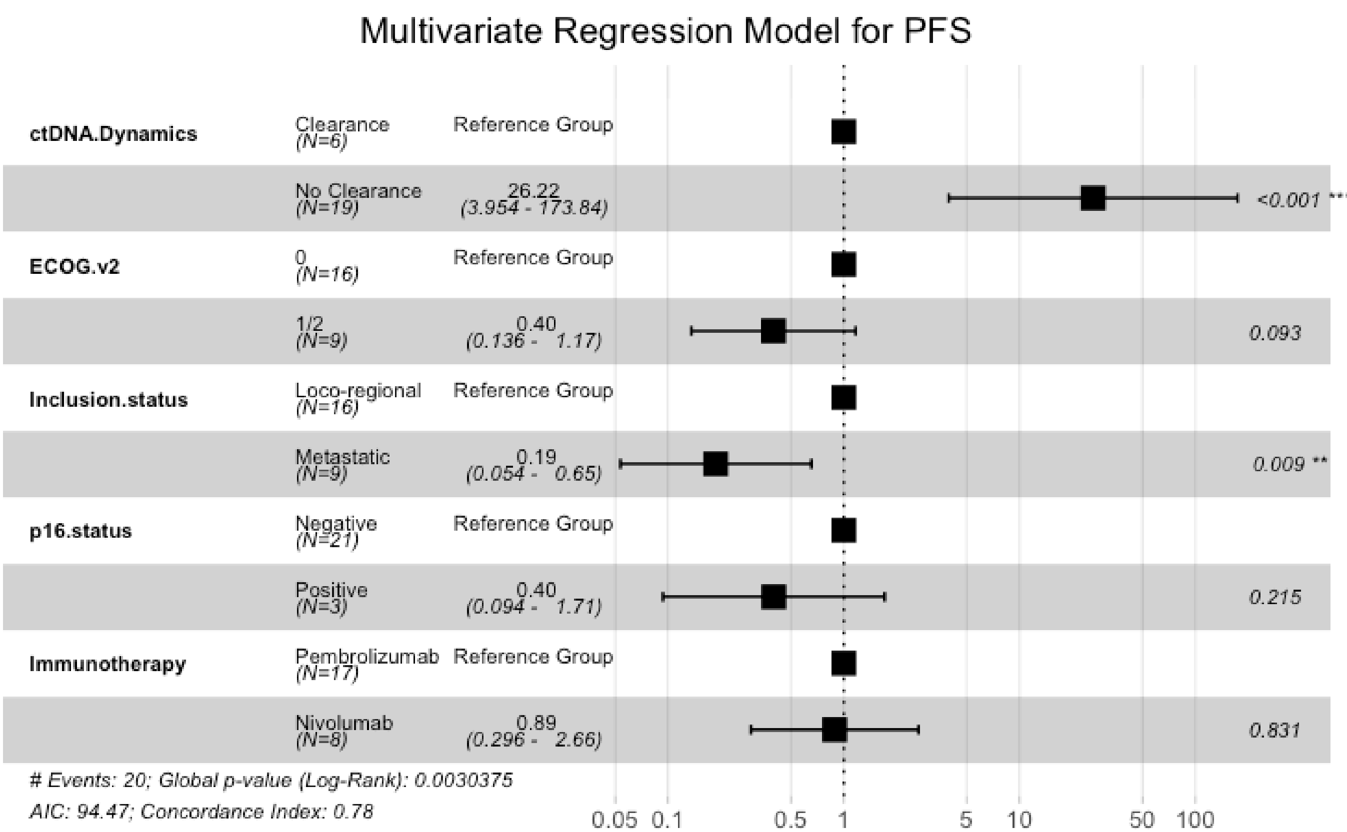
```
table_df <- as.data.frame(contingency_table)
table_df$Total <- ave(table_df$Freq, table_df$Var1, FUN = sum)
table_df$Percentage <- table_df$Freq / table_df$Total
table_df$MiddlePercentage <- table_df$Percentage / 2
ggplot(table_df, aes(x = Var1, y = Percentage, fill = Var2)) +
  geom_bar(stat = "identity") +
  geom_text(aes(y = MiddlePercentage, label = Freq), position = "stack", color = "blac
k", vjust = 1.5, size = 7) +
  theme_minimal() +
  labs(title = "ctDNA clearance post/under-treatment",
       x = "ctDNA",
       y = "Patients (%)",
       fill = "BOR",
       caption = paste("Fisher's exact test p-value: ", format.pval(fisher_exact_test$p.
value))) +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_manual(values = c("CR" = "blue", "PR" = "lightblue", "SD" = "orange", "PD"
= "red")) + # define custom colors
  theme(axis.text.x = element_text(angle = 0, hjust = 1.5, size = 14), # increase x-axis
text size
        axis.text.y = element_text(size = 14, color = "black"), # increase y-axis text s
ize
        axis.title.x = element_text(size = 14, color = "black"), # increase x-axis label
size
        axis.title.y = element_text(size = 14, color = "black"), # increase y-axis label
size
        legend.text = element_text(size = 12, color = "black"))  # increase Progression
label size
```



#Multivariate cox regression for PFS - ctDNA clearance

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data <- circ_data[circ_data$ctDNA.Base=="POSITIVE",]
circ_datadf <- as.data.frame(circ_data)

circ_data$ctDNA.Dynamics <- NA
circ_data <- circ_data %>%
  mutate(ctDNA.Dynamics = case_when(
    ctDNA.Base == "POSITIVE" & ctDNA.postTx == "NEGATIVE" ~ 1,
    ctDNA.Base == "POSITIVE" & ctDNA.postTx == "POSITIVE" ~ 2
  ))

circ_data$ctDNA.Dynamics <- factor(circ_data$ctDNA.Dynamics, levels=c("1","2"), labels = c("Clearance", "No Clearance"))
circ_data$Inclusion.status <- factor(circ_data$Inclusion.status, levels = c("Loco-regional", "Metastatic"))
circ_data$p16.status <- factor(circ_data$p16.status, levels = c("Negative", "Positive"))
circ_data$CPS.Scorev2 <- factor(circ_data$CPS.Scorev2, levels = c("≥1", "<1"))
circ_data$ECOG.v2 <- factor(circ_data$ECOG.v2, levels = c("0", "1/2"))
circ_data$Immunotherapy <- factor(circ_data$Immunotherapy, levels = c("Pembrolizumab", "Nivolumab"))
surv_object <- Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
cox_fit <- coxph(surv_object ~ ctDNA.Dynamics + ECOG.v2 + Inclusion.status + p16.status + Immunotherapy, data=circ_data)
ggforest(cox_fit, data = circ_data, main = "Multivariate Regression Model for PFS", refLabel = "Reference Group")
```



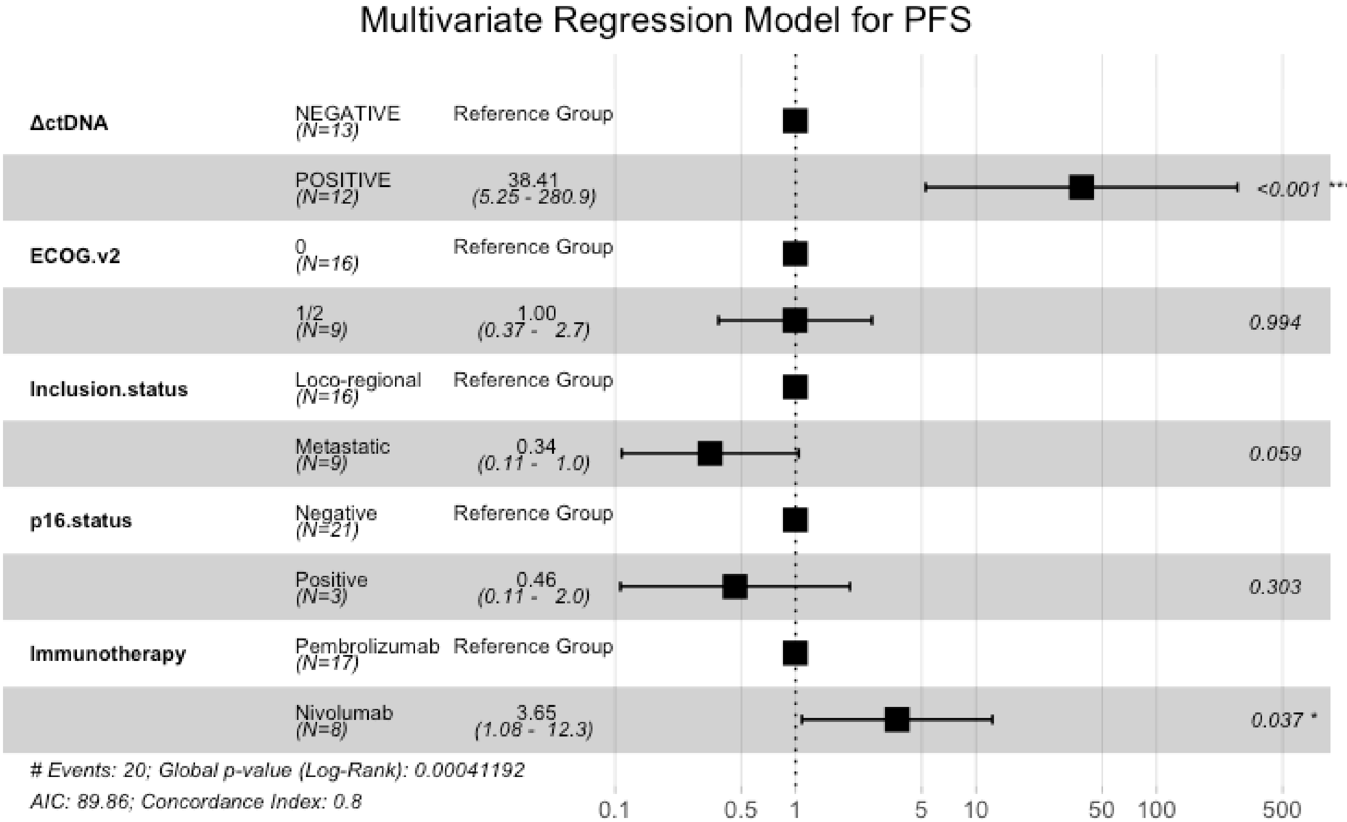
Hide

```
test.ph <- cox.zph(cox_fit)
```

#Multivariate cox regression for PFS - ctDNA kinetics

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ΔctDNA!="",]
circ_datadf <- as.data.frame(circ_data)

circ_data$ΔctDNA <- factor(circ_data$ΔctDNA, levels=c("NEGATIVE","POSITIVE"))
circ_data$Inclusion.status <- factor(circ_data$Inclusion.status, levels = c("Loco-region
al", "Metastatic"))
circ_data$p16.status <- factor(circ_data$p16.status, levels = c("Negative", "Positive"))
circ_data$CPS.Scorev2 <- factor(circ_data$CPS.Scorev2, levels = c("≥1", "<1"))
circ_data$ECOG.v2 <- factor(circ_data$ECOG.v2, levels = c("0", "1/2"))
circ_data$Immunotherapy <- factor(circ_data$Immunotherapy, levels = c("Pembrolizumab",
"Nivolumab"))
surv_object <- Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
cox_fit <- coxph(surv_object ~ ΔctDNA + ECOG.v2 + Inclusion.status + p16.status + Immuno
therapy, data=circ_data)
ggforest(cox_fit, data = circ_data, main = "Multivariate Regression Model for PFS", refL
abel = "Reference Group")
```



Hide

```
test.ph <- cox.zph(cox_fit)
```

#Univariate PFS cox regression for variables included in MVA

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data <- circ_data[circ_data$ctDNA.Base=="POSITIVE",]

circ_data$ctDNA.Dynamics <- NA
circ_data <- circ_data %>%
  mutate(ctDNA.Dynamics = case_when(
    ctDNA.Base == "POSITIVE" & ctDNA.postTx == "NEGATIVE" ~ 1,
    ctDNA.Base == "POSITIVE" & ctDNA.postTx == "POSITIVE" ~ 2
  ))
surv_object <-Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
circ_data$ctDNA.Dynamics <- factor(circ_data$ctDNA.Dynamics, levels=c("1","2"), labels =
c("Clearance", "No Clearance")) #univariate for ctDNA clearance post-treatment
cox_fit <- coxph(surv_object ~ ctDNA.Dynamics, data=circ_data)
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ ctDNA.Dynamics, data = circ_data)

  n= 25, number of events= 20

                              coef exp(coef) se(coef)     z Pr(>|z|)
ctDNA.DynamicsNo Clearance 2.085     8.048    0.767 2.719  0.00655 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

                           exp(coef) exp(-coef) lower .95 upper .95
ctDNA.DynamicsNo Clearance     8.048     0.1243      1.79     36.19

Concordance= 0.672  (se = 0.049 )
Likelihood ratio test= 11.61  on 1 df,   p=7e-04
Wald test            = 7.39  on 1 df,   p=0.007
Score (logrank) test = 9.78  on 1 df,   p=0.002
```

Hide

```
cox_fit_summary <- summary(cox_fit)

#Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_C
I, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 8.05 (1.79-36.19); p = 0.007"
```

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ΔctDNA!="",]
surv_object <-Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
circ_data$ΔctDNA <- factor(circ_data$ΔctDNA, levels=c("NEGATIVE","POSITIVE"))
cox_fit <- coxph(surv_object ~ ΔctDNA, data=circ_data) #univariate for ΔctDNA
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ ΔctDNA, data = circ_data)

  n= 25, number of events= 20

                 coef exp(coef) se(coef)    z Pr(>|z|)
ΔctDNAPOSITIVE  2.716    15.112    0.794 3.42 0.000626 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

               exp(coef) exp(-coef) lower .95 upper .95
ΔctDNAPOSITIVE     15.11    0.06617     3.188     71.64

Concordance= 0.716  (se = 0.044 )
Likelihood ratio test= 17.9  on 1 df,   p=2e-05
Wald test            = 11.7  on 1 df,   p=6e-04
Score (logrank) test = 18.33  on 1 df,   p=2e-05
```

Hide

```
cox_fit_summary <- summary(cox_fit)

#Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_C
I, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 15.11 (3.19-71.64); p = 0.001"
```

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data <- circ_data[circ_data$ctDNA.Base=="POSITIVE",]
surv_object <-Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
circ_data$ECOG.v2 <- factor(circ_data$ECOG.v2, levels = c("0", "1/2"))
cox_fit <- coxph(surv_object ~ ECOG.v2, data=circ_data) #univariate for ECOG
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ ECOG.v2, data = circ_data)

  n= 25, number of events= 20 

              coef exp(coef) se(coef)     z Pr(>|z|)
ECOG.v21/2 0.05209   1.05347  0.47029 0.111    0.912

           exp(coef) exp(-coef) lower .95 upper .95
ECOG.v21/2     1.053     0.9492    0.4191     2.648

Concordance= 0.521  (se = 0.062 )
Likelihood ratio test= 0.01  on 1 df,   p=0.9
Wald test            = 0.01  on 1 df,   p=0.9
Score (logrank) test = 0.01  on 1 df,   p=0.9
```

Hide

```
cox_fit_summary <- summary(cox_fit)

#Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_C
I, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 1.05 (0.42-2.65); p = 0.912"
```

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data <- circ_data[circ_data$ctDNA.Base=="POSITIVE",]
surv_object <-Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
circ_data$Inclusion.status <- factor(circ_data$Inclusion.status, levels = c("Loco-region
al", "Metastatic")) #univariate for Stage/disease status
cox_fit <- coxph(surv_object ~ Inclusion.status, data=circ_data)
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ Inclusion.status, data = circ_data)

  n= 25, number of events= 20

                              coef exp(coef) se(coef)      z Pr(>|z|)
Inclusion.statusMetastatic -0.1110    0.8949   0.4582 -0.242    0.809

                           exp(coef) exp(-coef) lower .95 upper .95
Inclusion.statusMetastatic    0.8949      1.117    0.3646     2.197

Concordance= 0.556  (se = 0.056 )
Likelihood ratio test= 0.06  on 1 df,   p=0.8
Wald test            = 0.06  on 1 df,   p=0.8
Score (logrank) test = 0.06  on 1 df,   p=0.8
```

Hide

```
cox_fit_summary <- summary(cox_fit)

#Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_C
I, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 0.89 (0.36-2.2); p = 0.809"
```

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data <- circ_data[circ_data$ctDNA.Base=="POSITIVE",]
surv_object <-Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
circ_data$p16.status <- factor(circ_data$p16.status, levels = c("Negative", "Positive"))
#univariate for p16 status
cox_fit <- coxph(surv_object ~ p16.status, data=circ_data)
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ p16.status, data = circ_data)

  n= 24, number of events= 20
   (1 observation deleted due to missingness)

                     coef exp(coef) se(coef)    z Pr(>|z|)
p16.statusPositive 0.2431    1.2752   0.6402 0.38    0.704

                   exp(coef) exp(-coef) lower .95 upper .95
p16.statusPositive     1.275     0.7842    0.3636     4.472

Concordance= 0.491  (se = 0.027 )
Likelihood ratio test= 0.14  on 1 df,   p=0.7
Wald test            = 0.14  on 1 df,   p=0.7
Score (logrank) test = 0.14  on 1 df,   p=0.7
```

Hide

```
cox_fit_summary <- summary(cox_fit)

#Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_C
I, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 1.28 (0.36-4.47); p = 0.704"
```

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data <- circ_data[circ_data$ctDNA.Base=="POSITIVE",]
surv_object <-Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
circ_data$Immunotherapy <- factor(circ_data$Immunotherapy, levels = c("Pembrolizumab",
"Nivolumab"))
cox_fit <- coxph(surv_object ~ Immunotherapy, data=circ_data) #univariate for Immunother
apy regimen
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ Immunotherapy, data = circ_data)

  n= 25, number of events= 20

                         coef exp(coef) se(coef)     z Pr(>|z|)
ImmunotherapyNivolumab 0.2901    1.3365   0.4709 0.616    0.538

                       exp(coef) exp(-coef) lower .95 upper .95
ImmunotherapyNivolumab     1.337     0.7482     0.531     3.364

Concordance= 0.54  (se = 0.061 )
Likelihood ratio test= 0.37  on 1 df,   p=0.5
Wald test            = 0.38  on 1 df,   p=0.5
Score (logrank) test = 0.38  on 1 df,   p=0.5
```

Hide

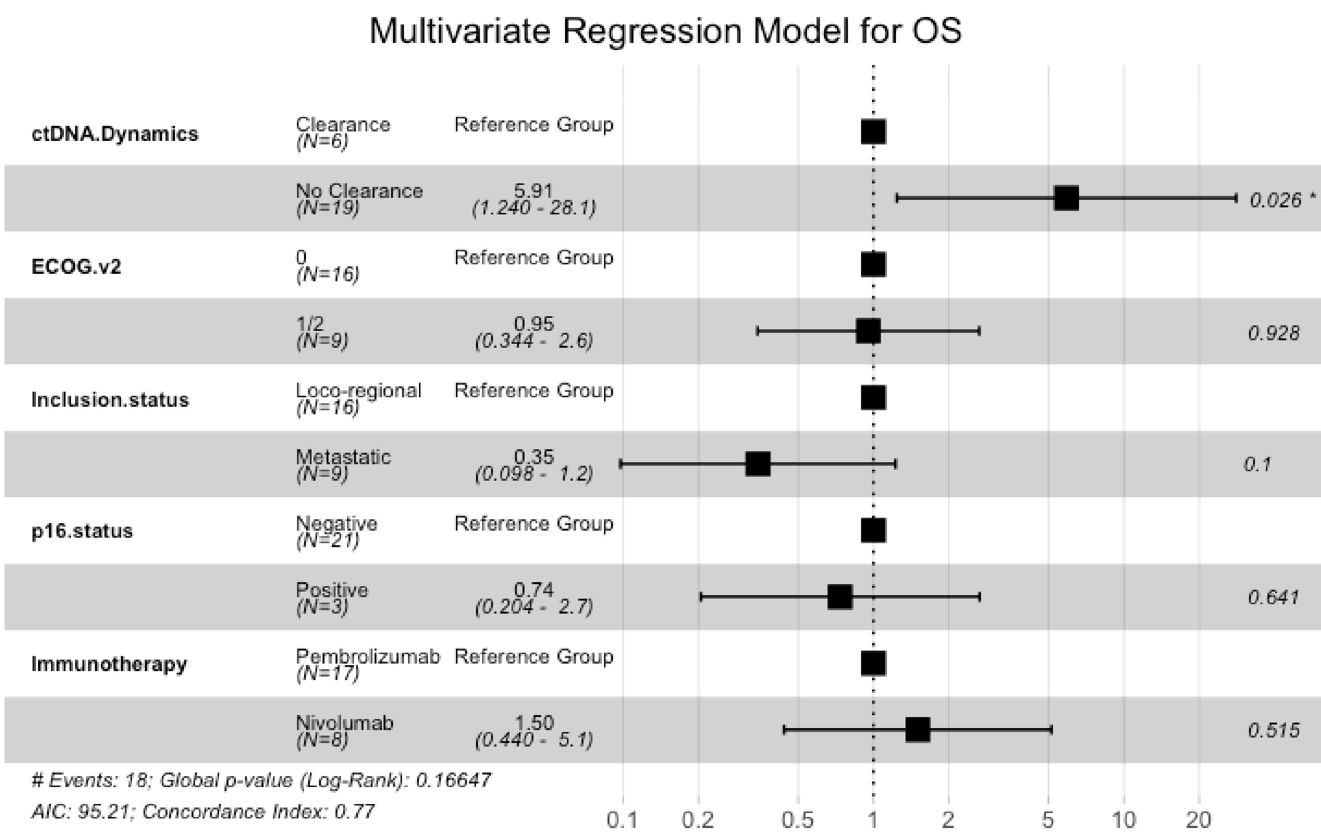
```
cox_fit_summary <- summary(cox_fit)

#Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_C
I, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 1.34 (0.53-3.36); p = 0.538"
```

#Multivariate cox regression for OS - ctDNA clearance

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data <- circ_data[circ_data$ctDNA.Base=="POSITIVE",]
circ_datadf <- as.data.frame(circ_data)

circ_data$ctDNA.Dynamics <- NA
circ_data <- circ_data %>%
  mutate(ctDNA.Dynamics = case_when(
    ctDNA.Base == "POSITIVE" & ctDNA.postTx == "NEGATIVE" ~ 1,
    ctDNA.Base == "POSITIVE" & ctDNA.postTx == "POSITIVE" ~ 2
  ))

circ_data$ctDNA.Dynamics <- factor(circ_data$ctDNA.Dynamics, levels=c("1","2"), labels =
c("Clearance", "No Clearance"))
circ_data$Inclusion.status <- factor(circ_data$Inclusion.status, levels = c("Loco-region
al", "Metastatic"))
circ_data$p16.status <- factor(circ_data$p16.status, levels = c("Negative", "Positive"))
circ_data$CPS.Scorev2 <- factor(circ_data$CPS.Scorev2, levels = c("≥1", "<1"))
circ_data$ECOG.v2 <- factor(circ_data$ECOG.v2, levels = c("0", "1/2"))
circ_data$Immunotherapy <- factor(circ_data$Immunotherapy, levels = c("Pembrolizumab",
"Nivolumab"))
surv_object <- Surv(time = circ_data$OS.months, event = circ_data$OS.Event)
cox_fit <- coxph(surv_object ~ ctDNA.Dynamics + ECOG.v2 + Inclusion.status + p16.status
+ Immunotherapy, data=circ_data)
ggforest(cox_fit, data = circ_data, main = "Multivariate Regression Model for OS", refLa
bel = "Reference Group")
```



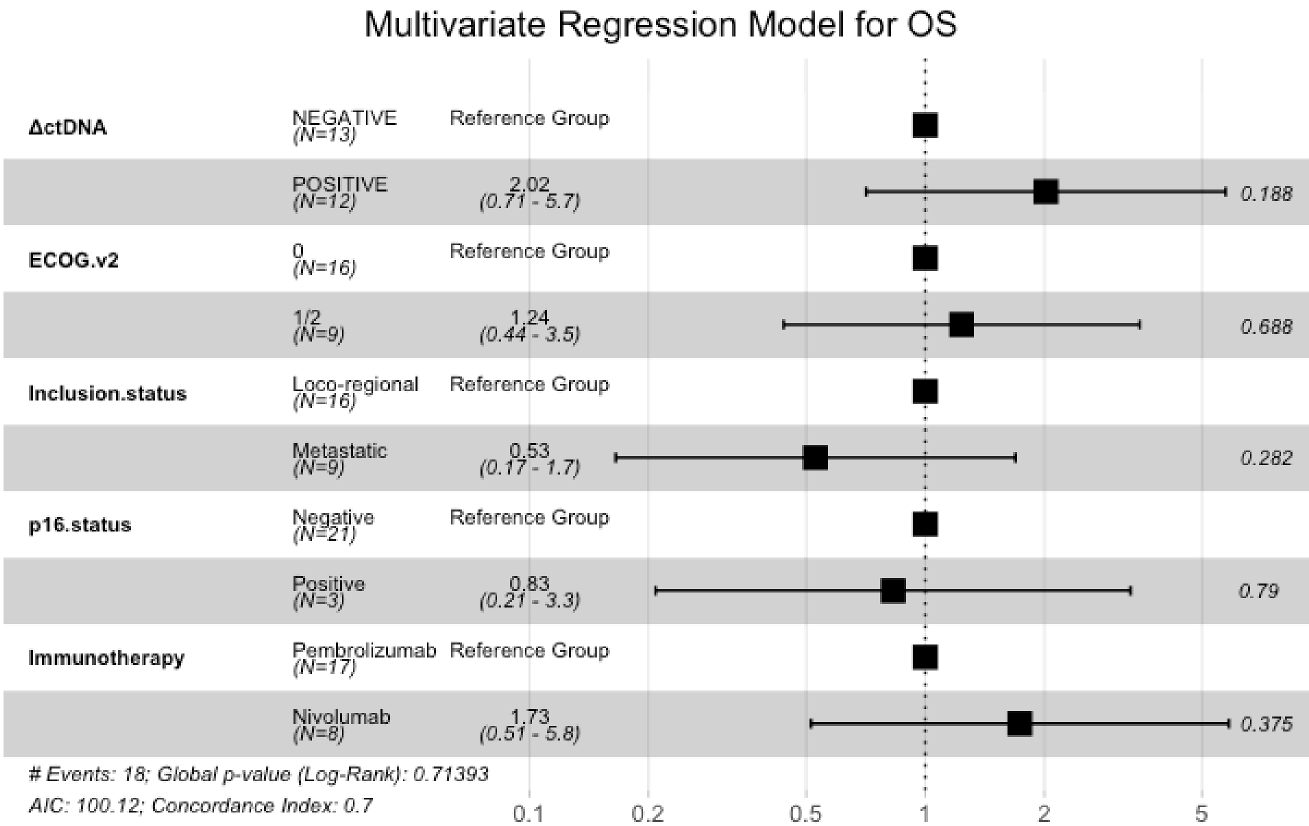
Hide

```
test.ph <- cox.zph(cox_fit)
```

#Multivariate cox regression for OS - ctDNA kinetics

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ΔctDNA!="",]
circ_datadf <- as.data.frame(circ_data)

circ_data$ΔctDNA <- factor(circ_data$ΔctDNA, levels=c("NEGATIVE","POSITIVE"))
circ_data$Inclusion.status <- factor(circ_data$Inclusion.status, levels = c("Loco-region
al", "Metastatic"))
circ_data$p16.status <- factor(circ_data$p16.status, levels = c("Negative", "Positive"))
circ_data$CPS.Scorev2 <- factor(circ_data$CPS.Scorev2, levels = c("≥1", "<1"))
circ_data$ECOG.v2 <- factor(circ_data$ECOG.v2, levels = c("0", "1/2"))
circ_data$Immunotherapy <- factor(circ_data$Immunotherapy, levels = c("Pembrolizumab",
"Nivolumab"))
surv_object <- Surv(time = circ_data$OS.months, event = circ_data$OS.Event)
cox_fit <- coxph(surv_object ~ ΔctDNA + ECOG.v2 + Inclusion.status + p16.status + Immuno
therapy, data=circ_data)
ggforest(cox_fit, data = circ_data, main = "Multivariate Regression Model for OS", refLa
bel = "Reference Group")
```



Hide

```
test.ph <- cox.zph(cox_fit)
```

#Univariate OS cox regression for variables included in MVA

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data <- circ_data[circ_data$ctDNA.Base=="POSITIVE",]

circ_data$ctDNA.Dynamics <- NA
circ_data <- circ_data %>%
  mutate(ctDNA.Dynamics = case_when(
    ctDNA.Base == "POSITIVE" & ctDNA.postTx == "NEGATIVE" ~ 1,
    ctDNA.Base == "POSITIVE" & ctDNA.postTx == "POSITIVE" ~ 2
  ))
surv_object <-Surv(time = circ_data$OS.months, event = circ_data$OS.Event)
circ_data$ctDNA.Dynamics <- factor(circ_data$ctDNA.Dynamics, levels=c("1","2"), labels =
c("Clearance", "No Clearance")) #univariate for ctDNA clearance post-treatment
cox_fit <- coxph(surv_object ~ ctDNA.Dynamics, data=circ_data)
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ ctDNA.Dynamics, data = circ_data)

  n= 25, number of events= 18

                             coef exp(coef) se(coef)     z Pr(>|z|)
ctDNA.DynamicsNo Clearance 1.6133    5.0196   0.7578 2.129   0.0333 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

                           exp(coef) exp(-coef) lower .95 upper .95
ctDNA.DynamicsNo Clearance      5.02     0.1992     1.137     22.17

Concordance= 0.65  (se = 0.047 )
Likelihood ratio test= 6.57  on 1 df,   p=0.01
Wald test            = 4.53  on 1 df,   p=0.03
Score (logrank) test = 5.52  on 1 df,   p=0.02
```

Hide

```
cox_fit_summary <- summary(cox_fit)

#Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_C
I, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 5.02 (1.14-22.17); p = 0.033"
```

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ΔctDNA!="",]
surv_object <-Surv(time = circ_data$OS.months, event = circ_data$OS.Event)
circ_data$ΔctDNA <- factor(circ_data$ΔctDNA, levels=c("NEGATIVE","POSITIVE"))
cox_fit <- coxph(surv_object ~ ΔctDNA, data=circ_data) #univariate for ΔctDNA
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ ΔctDNA, data = circ_data)

  n= 25, number of events= 18

                 coef exp(coef) se(coef)     z Pr(>|z|)
ΔctDNAPOSITIVE 0.6947    2.0031   0.4806 1.445    0.148

               exp(coef) exp(-coef) lower .95 upper .95
ΔctDNAPOSITIVE     2.003     0.4992    0.7809     5.138

Concordance= 0.598  (se = 0.061 )
Likelihood ratio test= 2.1  on 1 df,   p=0.1
Wald test            = 2.09  on 1 df,   p=0.1
Score (logrank) test = 2.17  on 1 df,   p=0.1
```

Hide

```
cox_fit_summary <- summary(cox_fit)

#Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_C
I, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 2 (0.78-5.14); p = 0.148"
```

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data <- circ_data[circ_data$ctDNA.Base=="POSITIVE",]
surv_object <-Surv(time = circ_data$OS.months, event = circ_data$OS.Event)
circ_data$ECOG.v2 <- factor(circ_data$ECOG.v2, levels = c("0", "1/2"))
cox_fit <- coxph(surv_object ~ ECOG.v2, data=circ_data) #univariate for ECOG
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ ECOG.v2, data = circ_data)

  n= 25, number of events= 18 

             coef exp(coef) se(coef)    z Pr(>|z|)
ECOG.v21/2 0.2417    1.2734   0.5040 0.48    0.632

           exp(coef) exp(-coef) lower .95 upper .95
ECOG.v21/2     1.273     0.7853    0.4742     3.419

Concordance= 0.562  (se = 0.065 )
Likelihood ratio test= 0.22  on 1 df,   p=0.6
Wald test            = 0.23  on 1 df,   p=0.6
Score (logrank) test = 0.23  on 1 df,   p=0.6
```

Hide

```
cox_fit_summary <- summary(cox_fit)

#Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_CI, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 1.27 (0.47-3.42); p = 0.632"
```

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data <- circ_data[circ_data$ctDNA.Base=="POSITIVE",]
surv_object <-Surv(time = circ_data$OS.months, event = circ_data$OS.Event)
circ_data$Inclusion.status <- factor(circ_data$Inclusion.status, levels = c("Loco-regional", "Metastatic")) #univariate for Stage/disease status
cox_fit <- coxph(surv_object ~ Inclusion.status, data=circ_data)
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ Inclusion.status, data = circ_data)

  n= 25, number of events= 18

                              coef exp(coef) se(coef)      z Pr(>|z|)
Inclusion.statusMetastatic -0.2314    0.7934   0.4852 -0.477    0.633

                           exp(coef) exp(-coef) lower .95 upper .95
Inclusion.statusMetastatic    0.7934       1.26    0.3065     2.054

Concordance= 0.583  (se = 0.055 )
Likelihood ratio test= 0.23  on 1 df,   p=0.6
Wald test            = 0.23  on 1 df,   p=0.6
Score (logrank) test = 0.23  on 1 df,   p=0.6
```

Hide

```
cox_fit_summary <- summary(cox_fit)

#Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_C
I, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 0.79 (0.31-2.05); p = 0.633"
```

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data <- circ_data[circ_data$ctDNA.Base=="POSITIVE",]
surv_object <-Surv(time = circ_data$OS.months, event = circ_data$OS.Event)
circ_data$p16.status <- factor(circ_data$p16.status, levels = c("Negative", "Positive"))
#univariate for p16 status
cox_fit <- coxph(surv_object ~ p16.status, data=circ_data)
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ p16.status, data = circ_data)

  n= 24, number of events= 18
   (1 observation deleted due to missingness)

                     coef exp(coef) se(coef)     z Pr(>|z|)
p16.statusPositive 0.1126    1.1192   0.6349 0.177    0.859

                   exp(coef) exp(-coef) lower .95 upper .95
p16.statusPositive     1.119     0.8935    0.3224     3.885

Concordance= 0.486  (se = 0.034 )
Likelihood ratio test= 0.03  on 1 df,   p=0.9
Wald test            = 0.03  on 1 df,   p=0.9
Score (logrank) test = 0.03  on 1 df,   p=0.9
```

Hide

```
cox_fit_summary <- summary(cox_fit)

#Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_C
I, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 1.12 (0.32-3.88); p = 0.859"
```

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data <- circ_data[circ_data$ctDNA.Base=="POSITIVE",]
surv_object <-Surv(time = circ_data$OS.months, event = circ_data$OS.Event)
circ_data$Immunotherapy <- factor(circ_data$Immunotherapy, levels = c("Pembrolizumab",
"Nivolumab"))
cox_fit <- coxph(surv_object ~ Immunotherapy, data=circ_data) #univariate for Immunother
apy regimen
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ Immunotherapy, data = circ_data)

  n= 25, number of events= 18

                          coef exp(coef) se(coef)     z Pr(>|z|)
ImmunotherapyNivolumab 0.2303    1.2590   0.4884 0.472    0.637

                       exp(coef) exp(-coef) lower .95 upper .95
ImmunotherapyNivolumab     1.259     0.7943    0.4834     3.279

Concordance= 0.517  (se = 0.065 )
Likelihood ratio test= 0.22  on 1 df,   p=0.6
Wald test            = 0.22  on 1 df,   p=0.6
Score (logrank) test = 0.22  on 1 df,   p=0.6
```

Hide

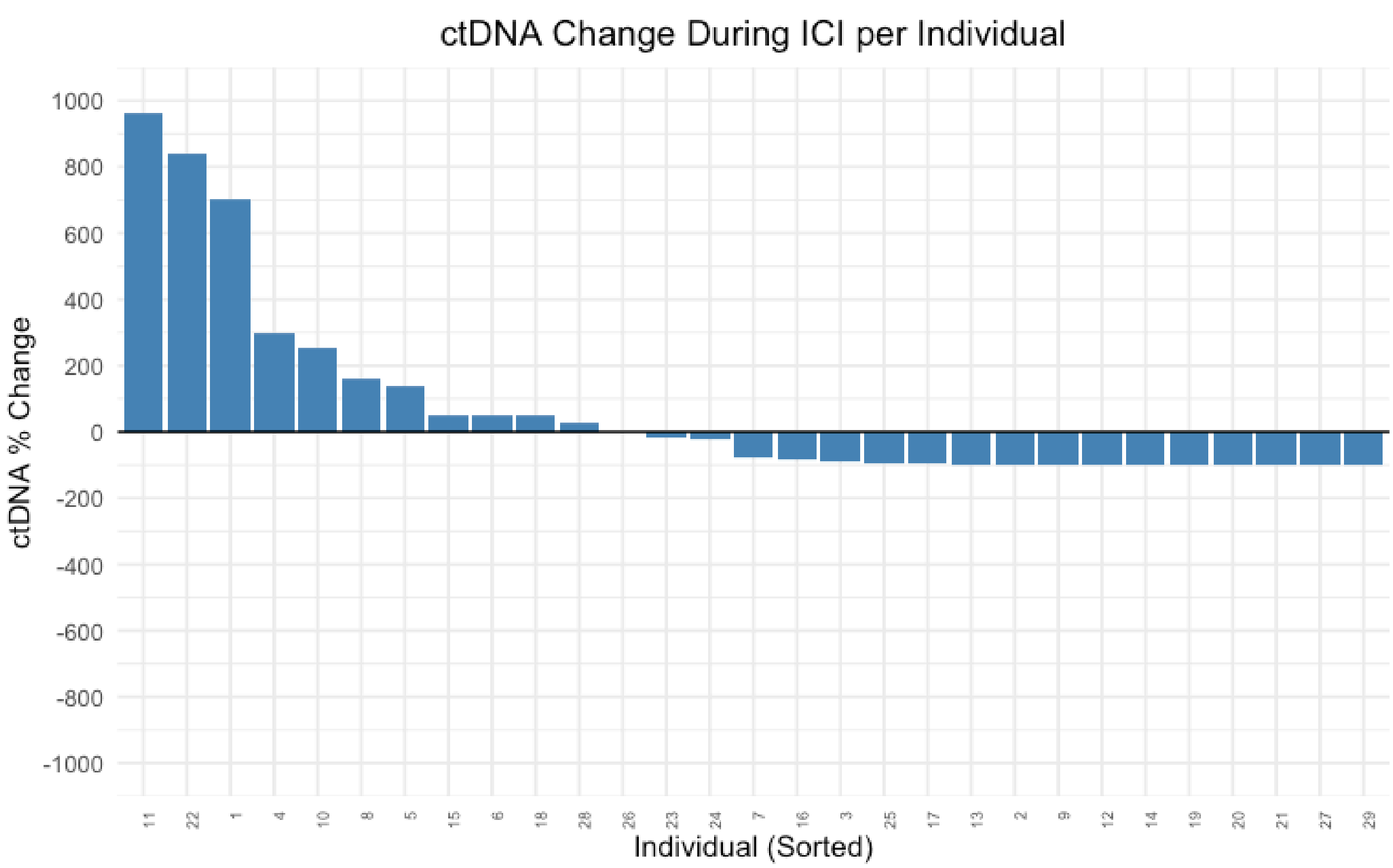
```
cox_fit_summary <- summary(cox_fit)

#Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_C
I, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 1.26 (0.48-3.28); p = 0.637"
```

#Waterfall with % change for MTM/mL

Hide

```
rm(list = ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available == "TRUE", ]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data$ctDNA.change <- as.numeric(circ_data$ctDNA.change)
circ_data$SubjectID <- seq_len(nrow(circ_data))
circ_data <- circ_data %>% arrange(desc(ctDNA.change))
circ_data$SubjectID <- factor(circ_data$SubjectID, levels = circ_data$SubjectID)

ggplot(circ_data, aes(x = SubjectID, y = ctDNA.change)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  geom_hline(yintercept = 0, color = "black") +
  scale_y_continuous(
    breaks = seq(-1000, 1000, by = 200),
    limits = c(-1000, 1000)
  ) +
  labs(
    x = "Individual (Sorted)",
    y = "ctDNA % Change",
    title = "ctDNA Change During ICI per Individual"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1, size = 6),
    plot.title = element_text(hjust = 0.5)
  )
```



#Heatmap for the clinical factors and top 10 most common mutations

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data$ctDNA.change <- as.numeric(circ_data$ctDNA.change)
circ_data <- circ_data %>% arrange(desc(ctDNA.change))
circ_datadf <- as.data.frame(circ_data)

ha <- HeatmapAnnotation(
  Inclusion.status = circ_data$Inclusion.status,
  Immunotherapy = circ_data$Immunotherapy,
  ECOG.v2 = circ_data$ECOG.v2,
  p16.status = circ_data$p16.status,
  TP53 = circ_data$TP53,
  CDKN2A = circ_data$CDKN2A,
  FAT1 = circ_data$FAT1,
  APOB = circ_data$APOB,
  LAMA1 = circ_data$LAMA1,
  NOTCH1 = circ_data$NOTCH1,
  PCDHB13 = circ_data$PCDHB13,
  DCHS1 = circ_data$DCHS1,
  FSIP2 = circ_data$FSIP2,
  LRP1B = circ_data$LRP1B,
  ctDNA.Base = circ_data$ctDNA.Base,
  ctDNA.postTx = circ_data$ctDNA.postTx,
  RESIST = circ_data$RESIST,
  PFS.Event = circ_data$PFS.Event,
  OS.Event = circ_data$OS.Event,

  col = list(Inclusion.status = c("Loco-regional" = "seagreen1", "Metastatic" = "khak
i"),
             Immunotherapy = c("Pembrolizumab" = "purple", "Nivolumab" ="lightblue"),
             ECOG.v2 = c("0" = "goldenrod" , "1/2" = "blue4"),
             p16.status = c("Negative" = "yellow", "Positive" ="brown", "Unknown" ="gre
y"),
             TP53 = c("MUT" = "black", "WT" ="grey"),
             CDKN2A = c("MUT" = "black", "WT" ="grey"),
             FAT1 = c("MUT" = "black", "WT" ="grey"),
             APOB = c("MUT" = "black", "WT" ="grey"),
             LAMA1 = c("MUT" = "black", "WT" ="grey"),
             NOTCH1 = c("MUT" = "black", "WT" ="grey"),
             PCDHB13 = c("MUT" = "black", "WT" ="grey"),
             DCHS1 = c("MUT" = "black", "WT" ="grey"),
             FSIP2 = c("MUT" = "black", "WT" ="grey"),
             LRP1B = c("MUT" = "black", "WT" ="grey"),
             ctDNA.Base = c("POSITIVE" = "red3", "NEGATIVE" ="blue"),
             ctDNA.postTx = c("POSITIVE" = "red3", "NEGATIVE" ="blue"),
             RESIST = c("CR" = "lightblue", "PR" ="blue", "SD" = "orange", "PD" ="red"),
             PFS.Event = c("TRUE" = "red3", "FALSE" ="blue"),
             OS.Event = c("TRUE" = "black", "FALSE" ="grey")
  )
```

```
)
ht <- Heatmap(matrix(nrow = 0, ncol = length(circ_data$Inclusion.status)),show_row_names
= FALSE,cluster_rows = F,cluster_columns = FALSE, top_annotation = ha)
pdf("heatmap.pdf",width = 7, height = 7)
draw(ht, annotation_legend_side = "bottom")
dev.off()
```

```
null device
          1
```

#Frequency of top 10 mutations in ctDNA clearance vs non-clearance

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("EORTC HNSCC ICI Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.available=="TRUE",]
circ_data <- circ_data[circ_data$ctDNA.Base!="",]
circ_data <- circ_data[circ_data$ctDNA.Base=="POSITIVE",]
circ_datadf <- as.data.frame(circ_data)

circ_data$ctDNA.Dynamics <- NA
circ_data <- circ_data %>%
  mutate(ctDNA.Dynamics = case_when(
    ctDNA.Base == "POSITIVE" & ctDNA.postTx == "NEGATIVE" ~ 1,
    ctDNA.Base == "POSITIVE" & ctDNA.postTx == "POSITIVE" ~ 2
  ))
circ_data$ctDNA.Dynamics <- factor(circ_data$ctDNA.Dynamics, levels=c("1","2"), labels =
c("Clearance", "No Clearance"))
circ_data$TP53 <- factor(circ_data$TP53, levels=c("WT","MUT"))
circ_data$CDKN2A <- factor(circ_data$CDKN2A, levels=c("WT","MUT"))
circ_data$FAT1 <- factor(circ_data$FAT1, levels=c("WT","MUT"))
circ_data$APOB <- factor(circ_data$APOB, levels=c("WT","MUT"))
circ_data$LAMA1 <- factor(circ_data$LAMA1, levels=c("WT","MUT"))
circ_data$NOTCH1 <- factor(circ_data$NOTCH1, levels=c("WT","MUT"))
circ_data$PCDHB13 <- factor(circ_data$CUBN, levels=c("WT","MUT"))
circ_data$DCHS1 <- factor(circ_data$DCHS1, levels=c("WT","MUT"))
circ_data$FSIP2 <- factor(circ_data$FSIP2, levels=c("WT","MUT"))
circ_data$LRP1B <- factor(circ_data$LRP1B, levels=c("WT","MUT"))

mutation_genes <- c("TP53", "CDKN2A", "FAT1", "APOB", "LAMA1", "NOTCH1", "PCDHB13", "DCH
S1", "FSIP2", "LRP1B")
summary_table <- data.frame(
  Gene = character(),
  Clearance_MUT_Percent = numeric(),
  No_Clearance_MUT_Percent = numeric(),
  Overall_MUT_Percent = numeric(),
  Fisher_p_value = numeric(),
  stringsAsFactors = FALSE
)
for (gene in mutation_genes) {
  subset_data <- circ_data[!is.na(circ_data[[gene]]) & !is.na(circ_data$ctDNA.Dynamics),
]
  tab <- table(subset_data[[gene]], subset_data$ctDNA.Dynamics)
  total_clearance <- sum(tab[, "Clearance"])
  total_no_clearance <- sum(tab[, "No Clearance"])
  total_all <- sum(tab)
  pct_clearance <- if ("MUT" %in% rownames(tab)) 100 * tab["MUT", "Clearance"] / total_c
learance else 0
  pct_no_clearance <- if ("MUT" %in% rownames(tab)) 100 * tab["MUT", "No Clearance"] / t
otal_no_clearance else 0
  pct_overall <- if ("MUT" %in% rownames(tab)) 100 * sum(tab["MUT", ]) / total_all else
0
  fisher_p <- if (all(dim(tab) == c(2,2))) fisher.test(tab)$p.value else NA
  summary_table <- rbind(summary_table, data.frame(
```

```
    Gene = gene,
    Clearance_MUT_Percent = round(pct_clearance, 1),
    No_Clearance_MUT_Percent = round(pct_no_clearance, 1),
    Overall_MUT_Percent = round(pct_overall, 1),
    Fisher_p_value = signif(fisher_p, 3)
  ))
}
print(summary_table)
```

Gene <chr>	Clearance_MUT_Percent <dbl>	No_Clearance_MUT_Percent <dbl>	Overall_MUT_Percent <dbl>	Fisher
TP53	83.3	73.7	76	
CDKN2A	66.7	26.3	36	
FAT1	33.3	21.1	24	
APOB	0.0	21.1	16	
LAMA1	16.7	15.8	16	
NOTCH1	33.3	10.5	16	
PCDHB13	16.7	15.8	16	
DCHS1	0.0	21.1	16	
FSIP2	16.7	15.8	16	
LRP1B	16.7	15.8	16	

1-10 of 10 rows